Statistical Tests

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1 Population claims, evidence and decisions

A method with QC target 50 µg/L gives a sample mean of 48.5 µg/L. Does it read systematically low, or could random variation explain the departure? Other questions compare upstream/downstream diclofenac concentrations or growth rates from independent algal cultures, as pictured in Fig. 1. A hypothesis describes a population parameter or distribution. It does not set our observed test statistic equal to a limit. The reasoning diagram Fig. 2 separates the population claim from the statistic used to check it. In the classical parameter comparisons taught here, H_0 contains equality and H_1 excludes it; this is not a universal definition of every possible hypothesis.

Fig. 1. Three illustrative measurement questions, not experimental findings.

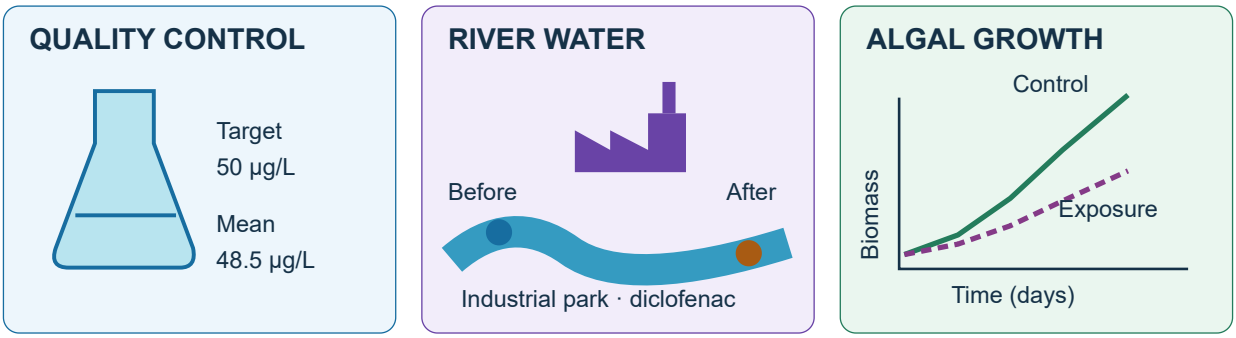
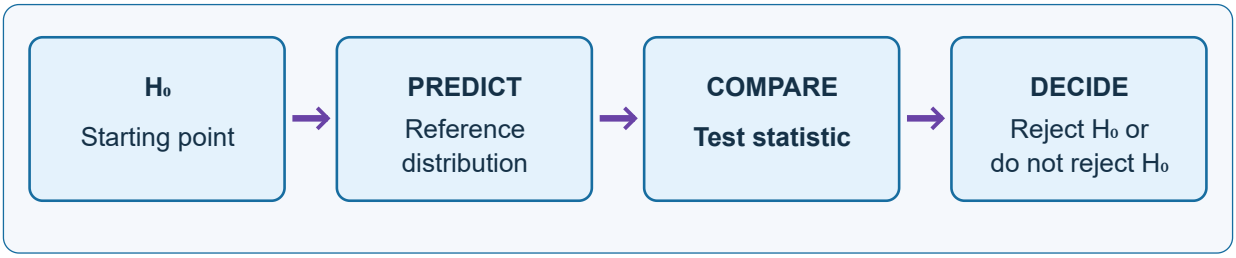
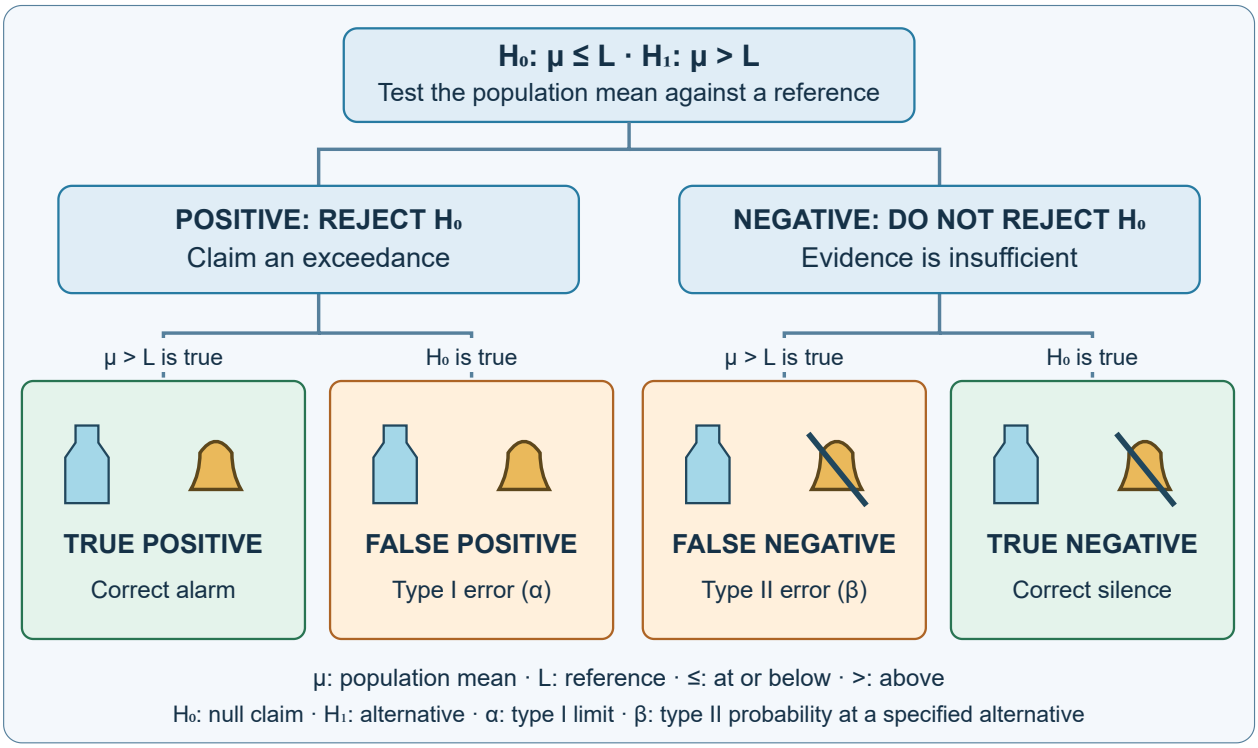


Fig. 2. From population claim to reference prediction, statistic and fallible decision.



The single tree in Fig. 3 starts with $\mathbf{H_0}$**:** $\mu \leq \mathbf{L}$, where μ is the population mean and L is the reference. **Positive** means rejecting H_0 and claiming an exceedance; **negative** means not rejecting H_0. The thumbnails distinguish true positive, false positive (type I), false negative (type II), and true negative. Non-rejection cannot prove H_0. The three lake hypotheses serve different applications. An equality-only GO policy treats thicker ice as a departure too; non-rejection cannot establish exact equality. Match the burden of evidence to the consequences of an error.

Fig. 3. One tree with positive/negative decisions and all four outcomes.



The lake illustration Fig. 4 is a thought experiment about that burden of evidence, not crossing advice. Rule A starts with thickness at or below a hypothetical minimum and goes only after rejection. Rule B starts with thickness at or above the minimum and goes on non-rejection. For the same known-SD Normal measurement model, the decision probabilities in Fig. 5 differ sharply. With four independent readings, population SD 0.5 cm and a 10 cm boundary, sample-mean SE is 0.25 cm. One-sided alpha 5% gives GO thresholds 10.411 cm for A and 9.589 cm for B. At true 9.7 cm, GO occurs about 0.22% versus 67.18%; at true 10.3 cm, about 32.82% versus 99.78%. These compare repeated studies at the same true state, not two analyses producing different physical collapse probabilities from one observed mean. Real ice has local weak spots and other hazards not represented by this model.

Fig. 4. Lake decision rules in a statistical thought experiment.

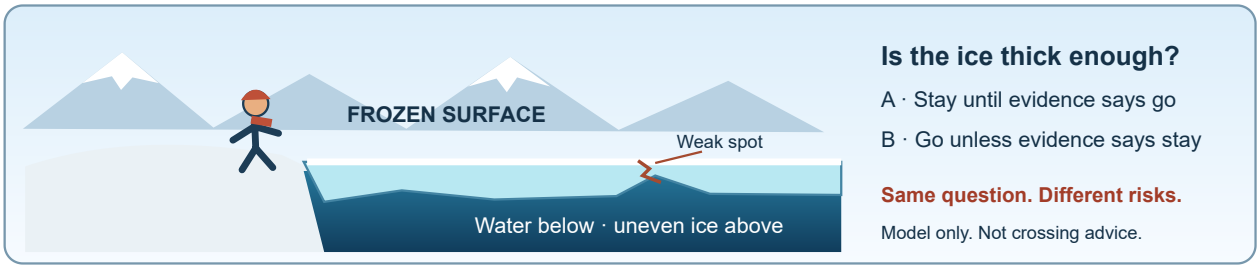
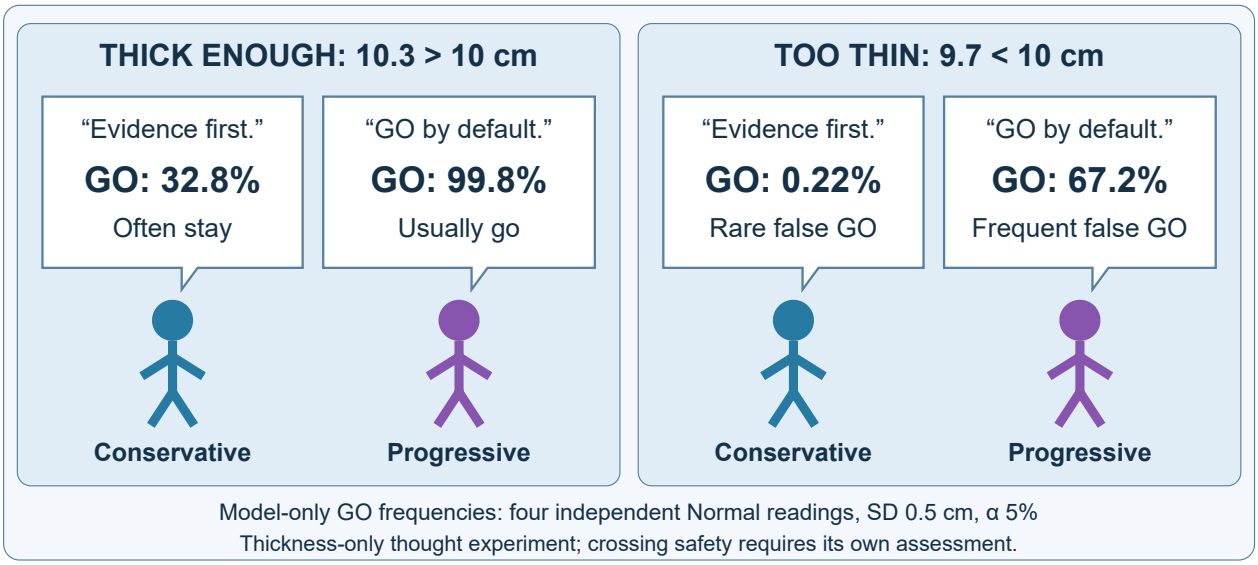


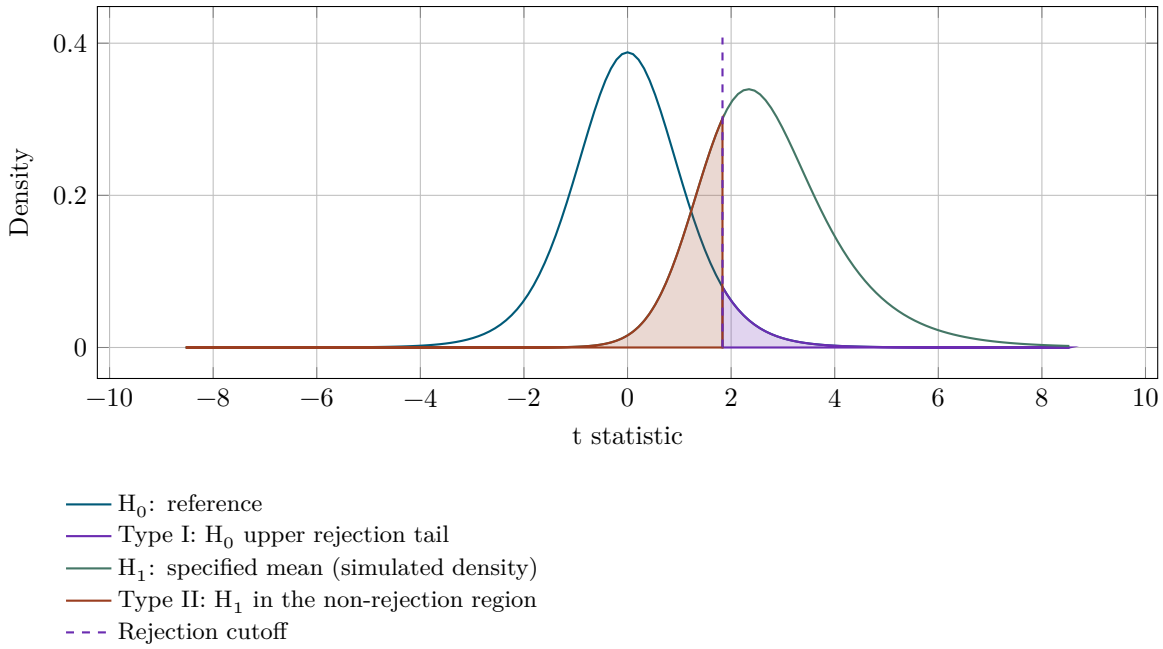
Fig. 5. Comic: Normal-model GO probabilities at two true thicknesses.



2 Error levels, degrees of freedom and p-values

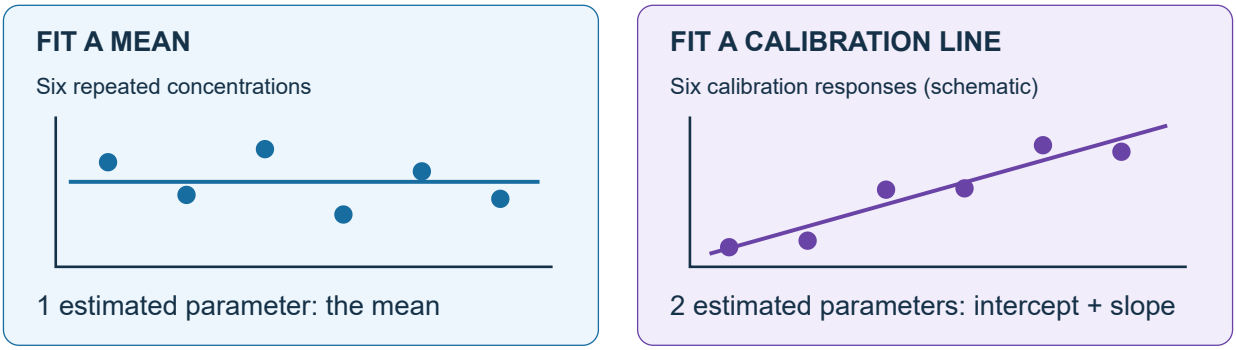
Choose alpha before observing the relevant outcomes. It bounds the procedure's false-rejection probability under H_0 and the sampling model. Lower alpha is not a free improvement: with the same number of observations, variability and true departure, it can increase missed differences. The interactive amoxicillin example Fig. 6 reuses 4000 seeded Normal-sample studies at a fixed true mean 1.4 µg/L versus teaching limit 1 µg/L. Ten independent results and population SD 0.5 µg/L define the model. The readouts distinguish worst-case type I error from simulated type II error.

Fig. 6. Alpha slider: a reference curve and type I/type II readouts; simulation settings are fixed.



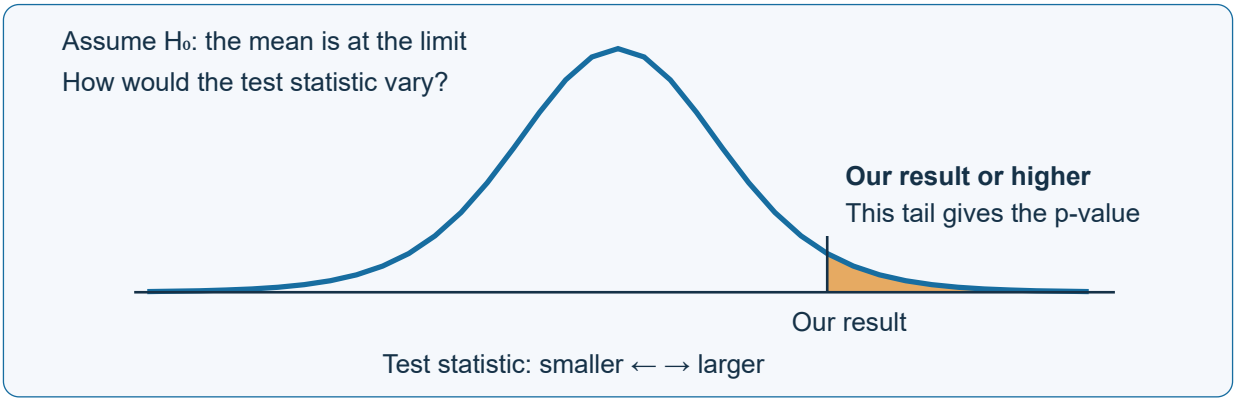
Degrees of freedom, written ν and pronounced nu, describe information remaining after fitting constraints. The residuals add to zero once their mean is fitted. Six independent observations therefore leave $\nu = \mathbf{5}$ for an SD estimate. Fitting intercept and slope leaves $\nu = \mathbf{4}$, as illustrated in Fig. 7. In these full-rank models, each independently fitted coefficient costs one residual degree of freedom. Other references can use a fractional approximation.

Fig. 7. Six observations with one fitted mean versus two fitted line coefficients.



A p-value is a probability conditional on H_0 and the sampling model: a statistic at least as extreme as the one observed, in the planned direction. The upper tail in Fig. 8 counts unusually high results for an exceedance question, with the null boundary used for calibration. Compare p with the alpha chosen in advance. Small p challenges the null model; it is not the probability that H_0 is true, a measurement of effect size, or evidence of causal mechanism.

Fig. 8. An observed upper-tail statistic and results at least as extreme under the null boundary.



3 A mean against a fixed target

The t-test section marker Fig. 9 introduces comparisons of means. A fixed-target comparison differs from comparing two estimated means, shown in Fig. 10. The **test statistic** Eq. 1 divides the sample mean's departure from its reference by estimated standard error. **SD** describes individual measurement scatter; **SE** describes sampling variation of the mean. Under independent Normal sampling with positive variance and a fixed null mean, the reference has $\nu = \mathbf{n} - \mathbf{1}$.

Fig. 9. The mean-comparison theme of t-tests.

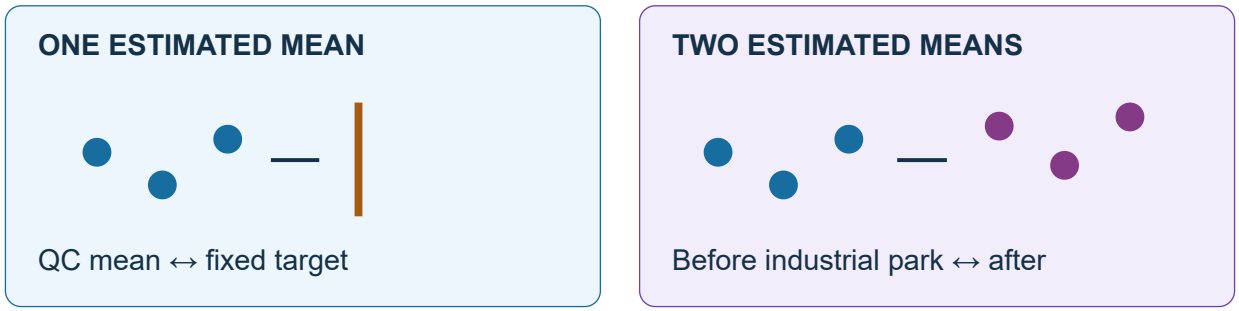
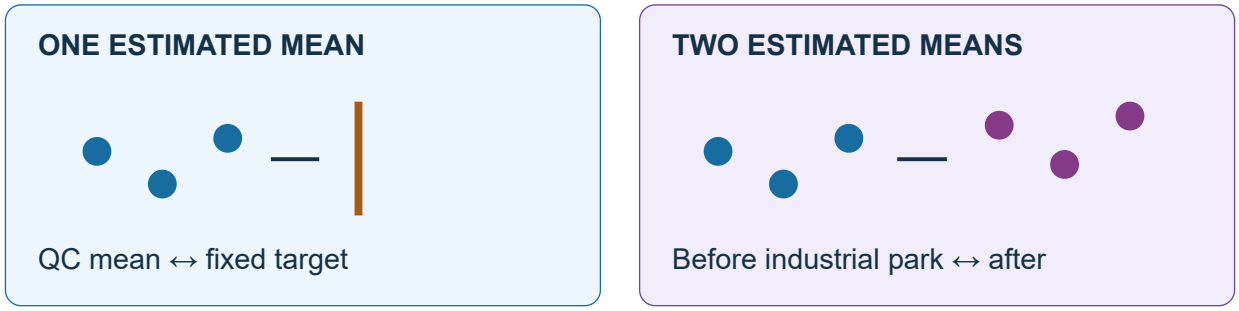


Fig. 10. One fixed reference or two estimated group means.



Eq. 1. One-sample t, SE and residual ν with complete symbols.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \nu = n - 1$$

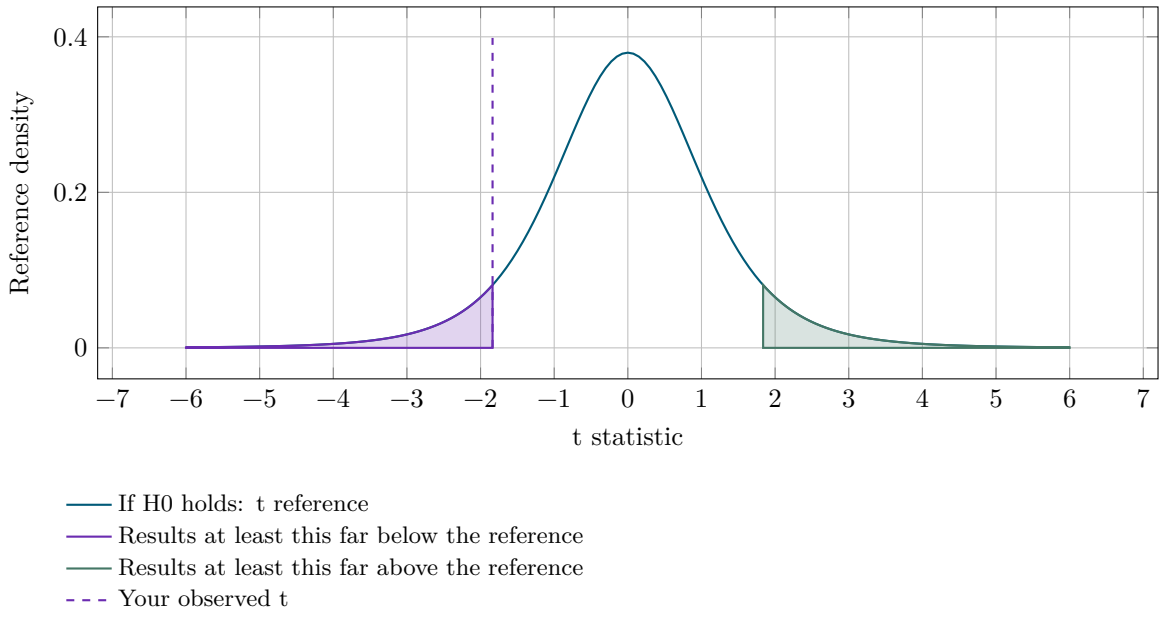
Symbols

- **t**: Observed test statistic; dimensionless.
- $\bar{x}$: Arithmetic sample mean; response unit.
- μ_0: Fixed null-reference population mean; same unit.
- **s**: Sample SD, using denominator n − 1; same unit.
- **n**: Number of independent observations; greater than one.
- ν **(nu)**: Reference degrees of freedom.
- $\sqrt{}$: Positive square root.
- $-$, /, $=$: Subtraction, division and equality.

Conditions: independent identically distributed Normal observations, positive variance and fixed reference. Juxtaposition multiplies; subscripts label quantities. Estimated reference uncertainty requires its own model.

The QC experiment Fig. 11 starts at target 50 µg/L, sample mean 48.5 µg/L, sample SD 2 µg/L and six independent preparations. It gives t about −1.837 and two-sided p about 0.126. Changing mean, SD or independent sample size illustrates hypothetical studies, not permission to edit observations or keep collecting until significance appears. The two-sided question counts both extremes.

Fig. 11. QC mean, SD and n explorer with its authored starting state.



The separate recovery example Fig. 12 asks whether a minimum population mean recovery of 80% is exceeded. $\mathbf{H_0}$**:** $\mu \leq$ **80%;** $\mathbf{H_1}$**:** $\mu >$ **80%**. The six synthetic values average **81.5%**. The R-style mimic Table 1 reports $\mathbf{t = 3.5032}$**,** ν **= 5, upper-tail p = 0.008612**. Rejection at 5% supports the population mean being above the illustrative minimum. Individual preparations still vary around that mean. Further reading: NIST one-sample t-test Fig. 13 and t reference table Fig. 14.

Fig. 12. Six synthetic recovery preparations, including separate repeated values.

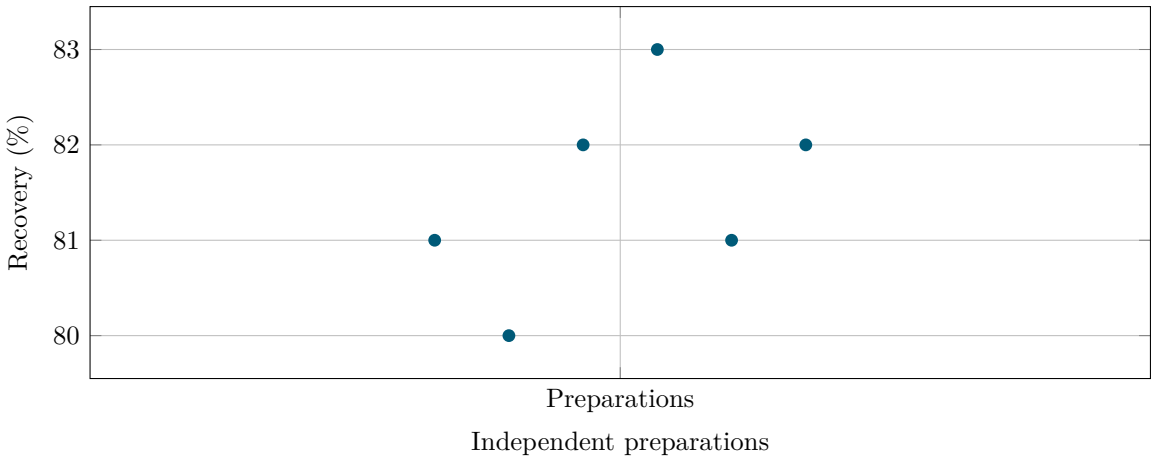


Table 1. R-style one-sided recovery report; authored mimic, not claimed live output.

Field	Result
Hypotheses	H_0: mean at or below 80%; H_1: above 80%
Statistic / ν	t = 3.5032; $\nu = 5$
p-value	0.008612 · upper t tail
Sample estimate	Mean 81.5%; n = 6; SD 1.049 percentage points
Reference	Upper t tail at the 80% null boundary
Decision at 5%	Reject H_0 under independent Normal preparations: evidence for a population mean above 80%. Individual preparations vary.

Fig. 13. NIST: one-sample t reference.

NIST: one-sample t-test

`https://www.itl.nist.gov/div898/handbook/eda/section3/eda352.htm`

Fig. 14. NIST: Student t critical values.

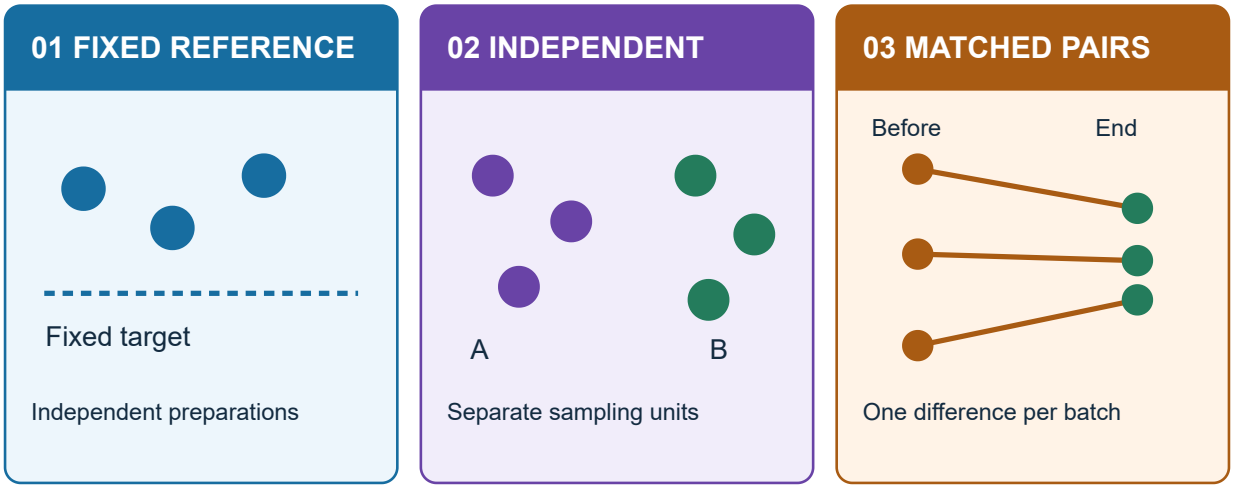
NIST: t reference table

`https://www.itl.nist.gov/div898/handbook/eda/section3/eda3672.htm`

4 Independent groups, shared variance and real pairs

Sampling identity comes before software choice. The structures in Fig. 15 distinguish independent groups from matched pairs. For independent Normal samples with a justified common variance, the pooled construction Eq. 2 combines sample variances using **residual degrees of freedom as weights**. The river drawing Fig. 16 represents separate upstream and downstream sites. Prior method validation and a homogeneous matrix could support similar within-region measurement variance. Another plausible setting is two randomized groups measured by the same validated assay.

Fig. 15. Fixed target, separate units and matched identities.



Eq. 2. Pooled variance weighted by residual degrees of freedom.

$$\underbrace{s_p^2}_{\text{weighted variance}} = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{s_p\sqrt{1/n_1+1/n_2}}$$

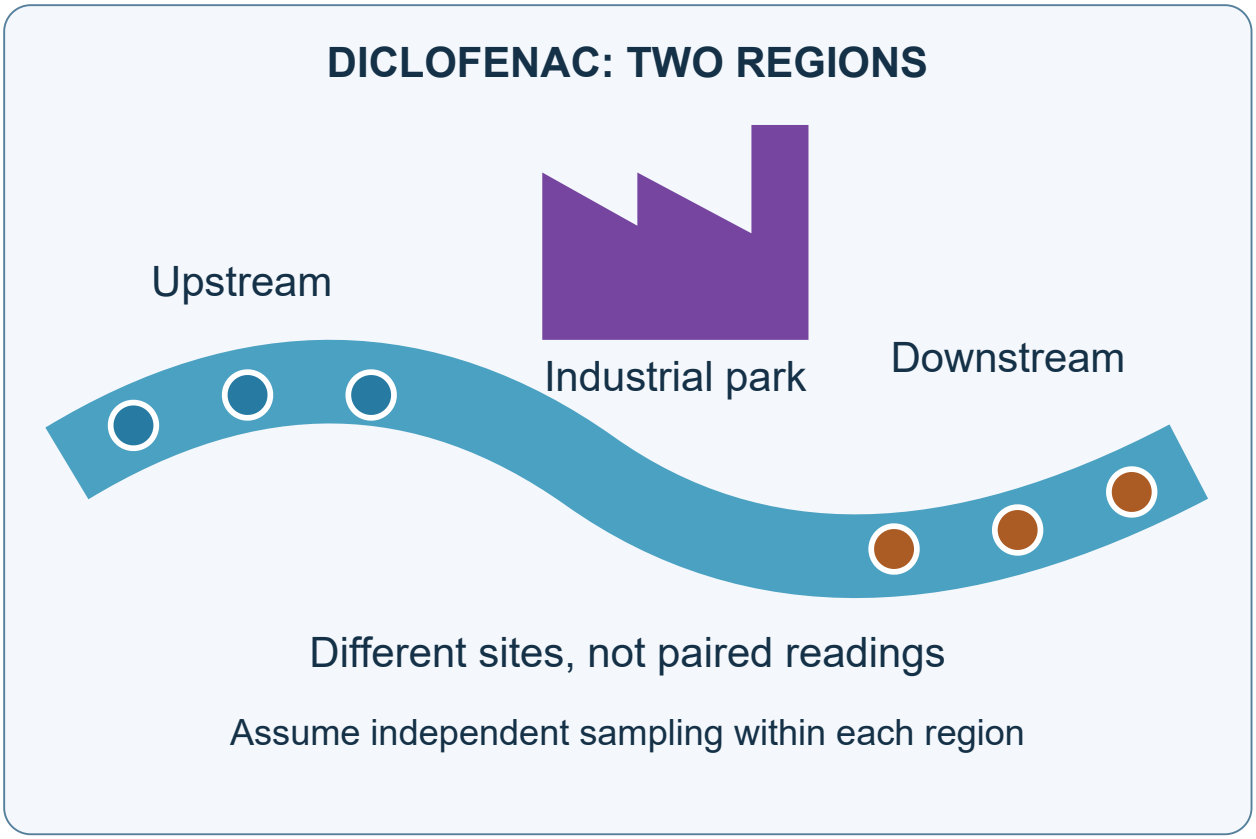
$$\nu = n_1 + n_2 - 2$$

Symbols

- **1, 2**: Independent-group indices.
- $\mathbf{n}_1$**,** $\mathbf{n}_2$: Group sample sizes, each greater than one.
- $\bar{x}_1$**,** $\bar{x}_2$: Group means; response unit.
- $\mathbf{s}_1{}^2$**,** $\mathbf{s}_2{}^2$: Sample variances, denominators $\mathrm{n}_1 - 1$ and $\mathrm{n}_2 - 1$; squared response unit.
- $\mathbf{s}_p{}^2$: Pooled variance weighted by residual degrees of freedom; p labels pooling.
- $\mathbf{s}_p$: Positive square root of pooled variance; response unit.
- $\boldsymbol{\Delta}_0$: Null population mean difference, often zero; response unit.
- **t;** ν: Dimensionless test statistic; residual degrees of freedom.
- **2;** $\sqrt{}$: Squaring; positive square root.
- **+, −, =, fraction bars**: Add, subtract, equal, divide.

Conditions: independent Normal samples with one common positive population variance. Juxtaposition multiplies; parentheses group. The underbrace labels the weighted variance.

Fig. 16. Independent upstream/downstream sampling around an industrial park.



An oxidation batch has a different structure. The before/end bottles in Fig. 17 share a batch, so keep that identity and form one **difference** per independent batch using Eq. 3. Exact paired-t inference assumes independent Normal differences. The six differences in Fig. 18 include zero and average **2.833 mg/L**. The report Table 2 gives $\mathbf{t = 4.0294}$, $\boldsymbol{\nu = 5}$, **two-sided** $\mathbf{p = 0.01003}$. All batches remain; the conclusion concerns the population mean before-minus-end decrease. Time-related competing explanations still matter for causal attribution.

Eq. 3. Form original batch differences before calculating a one-sample t statistic.

$$d_i = x_i - y_i, \quad t = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}, \quad \nu = n - 1$$

Symbols

- **i**: Original matched batch index.
- $\mathbf{x}_i$, $\mathbf{y}_i$: Before and end concentration in that same batch; mg/L.
- $\mathbf{d}_i$; $\bar{d}$: Before-minus-end difference; mean of differences; mg/L.
- **s_d**: Sample SD of differences, denominator n − 1; mg/L.
- $\boldsymbol{\Delta}_0$: Null mean difference, here zero; mg/L.
- **n**: Independent batches, not individual readings; greater than one.
- **t**; ν: Dimensionless statistic; reference degrees of freedom.
- $\sqrt{}$; −; =; **fraction bars**: Positive square root, subtract, equal, divide.

Conditions: independent Normal batch differences with positive variance. Original readings within a batch may be dependent; do not sort them to create new pairs. Subscripts label quantities; parentheses group.

Fig. 17. Before/end samples from one stirred oxidation batch are dependent.

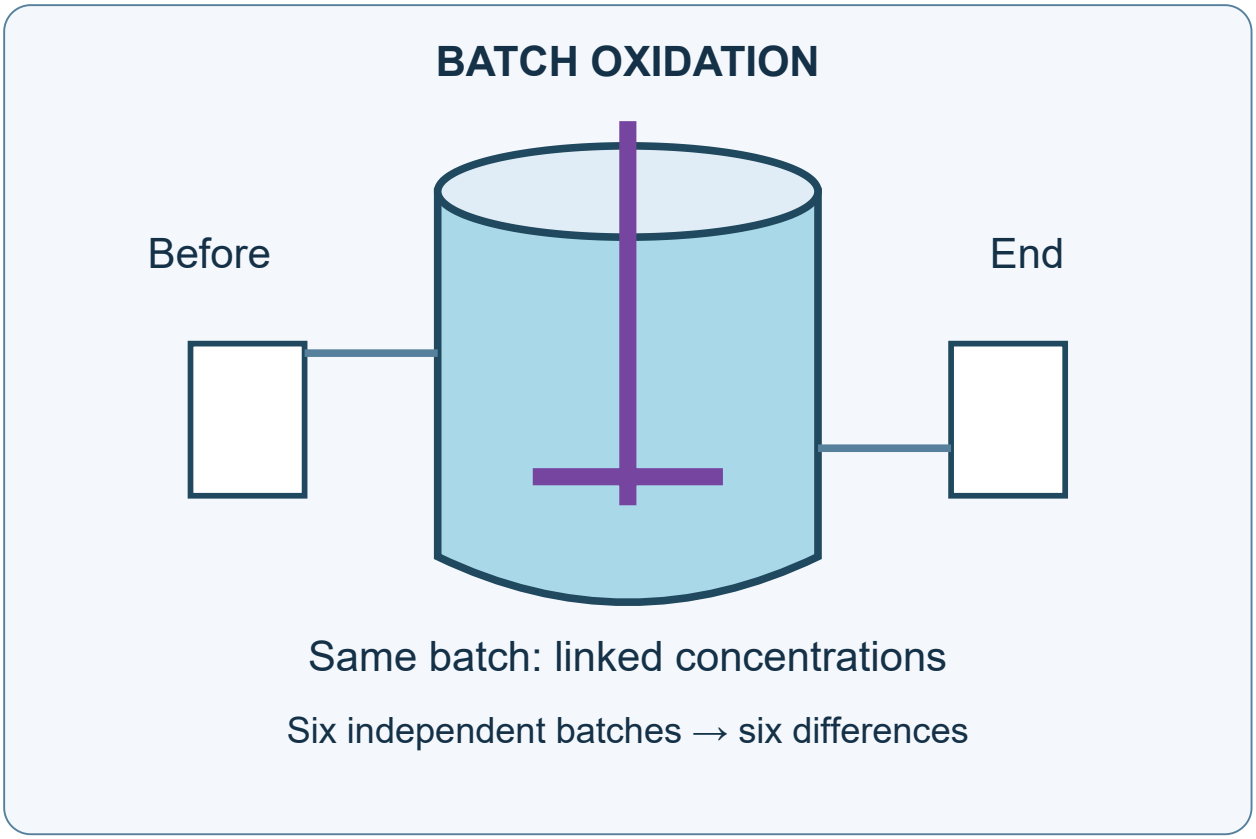
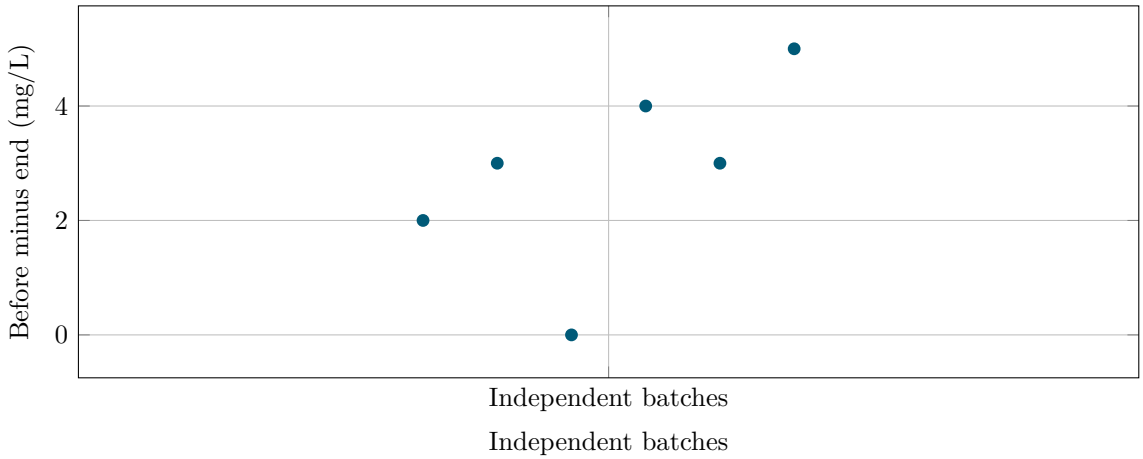


Fig. 18. Independent synthetic batch differences, including no observed change.



Independent batches

Table 2. Paired-t R-style report with the native-unit mean difference.

Field	Result
Alternative	H_0: zero population mean change; H_1: nonzero
Statistic / ν / p	t = 4.0294; $\nu = 5$; two-sided p = 0.01003
Mean difference	Before minus end: 2.833 mg/L; six batches
Reference	Two t tails under zero population mean difference
Decision / model	Reject at 5%; independent Normal batch differences. Mean concentration falls; control time-related alternatives before claiming causality.

Welch's construction Eq. 4 retains each group's own variance estimate and calibrates an approximate, generally fractional ν. The independent sets in Fig. 19 comprise five NIR and six HPLC-MS preparations from one homogeneous reference material. Shared samples measured twice would require pairing. The dot plot Fig. 20 has means **10 and 15 µg/L**. In Table 3, NIR minus HPLC-MS is **−5 µg/L**, with $\nu \approx$ **6.972, p** $\approx$ **0.02089**. Accuracy and method agreement require their own validation. NIST compares the three t constructions in Fig. 21.

Eq. 4. Separate mean-variance contributions and approximate Welch ν.

$$v_j = \frac{s_j^2}{n_j}, \quad t = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{v_1 + v_2}}$$
$$\nu = \frac{(v_1 + v_2)^2}{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)}$$

Symbols

- **j; 1, 2**: System index; NIR and HPLC-MS.
- **n_j**: Independent preparations in system j; greater than one.
- **s_j**^{2}: Sample variance, denominator n_j − 1; squared µg/L.
- **v_j**: Estimated variance of the system mean; squared µg/L.
- $\bar{x}_1$, $\bar{x}_2$; $\boldsymbol{\Delta}_0$: System means; null mean difference; µg/L.
- **t;** ν: Dimensionless statistic; approximate, possibly fractional degrees of freedom.
- **2;** $\sqrt{}$: Squaring; positive square root.
- **+, −, =, fraction bars**: Add, subtract, equal, divide.

Conditions: independent Normal groups, unequal positive variances allowed. Reference is approximate. Juxtaposition multiplies; parentheses group; subscripts identify independent systems.

Fig. 19. Independent NIR and HPLC-MS preparations, not a paired sample comparison.

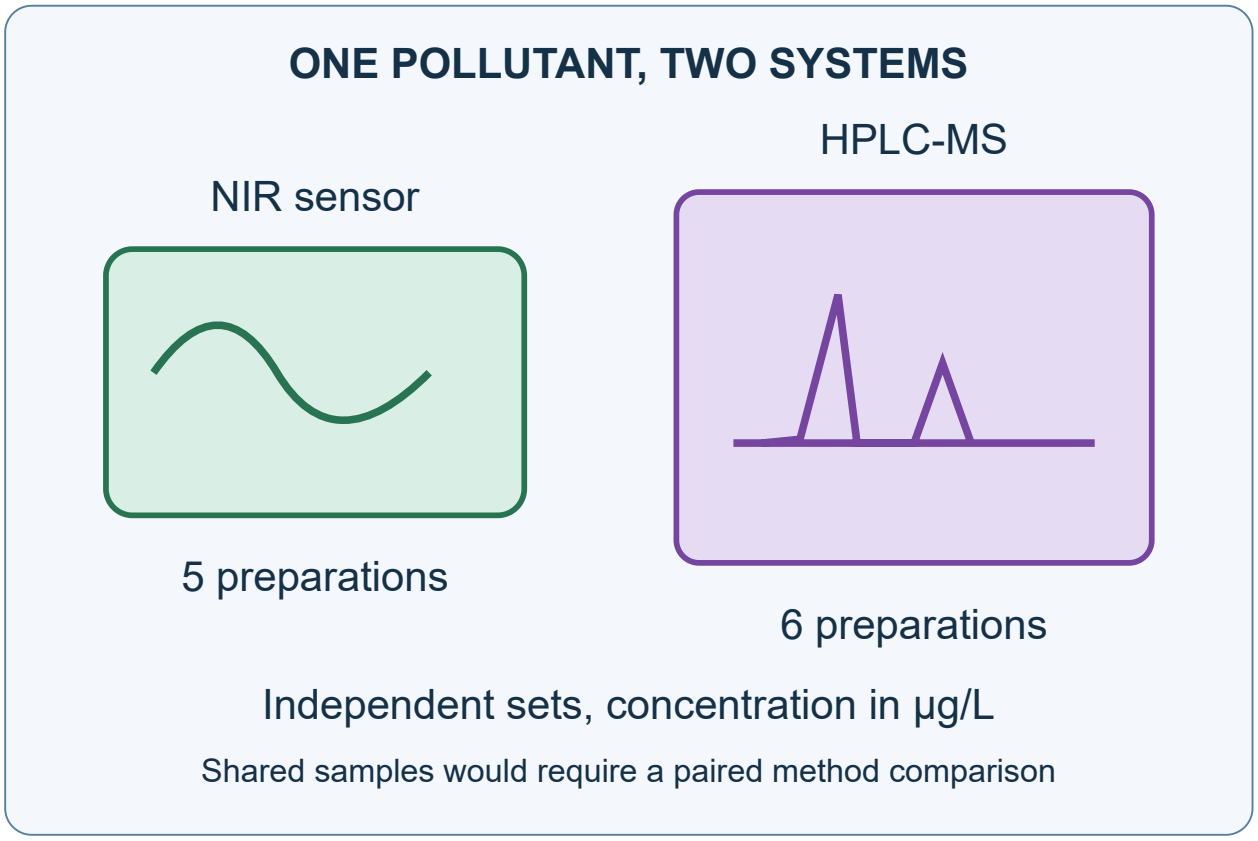


Fig. 20. Every independent concentration observation remains in its system group.

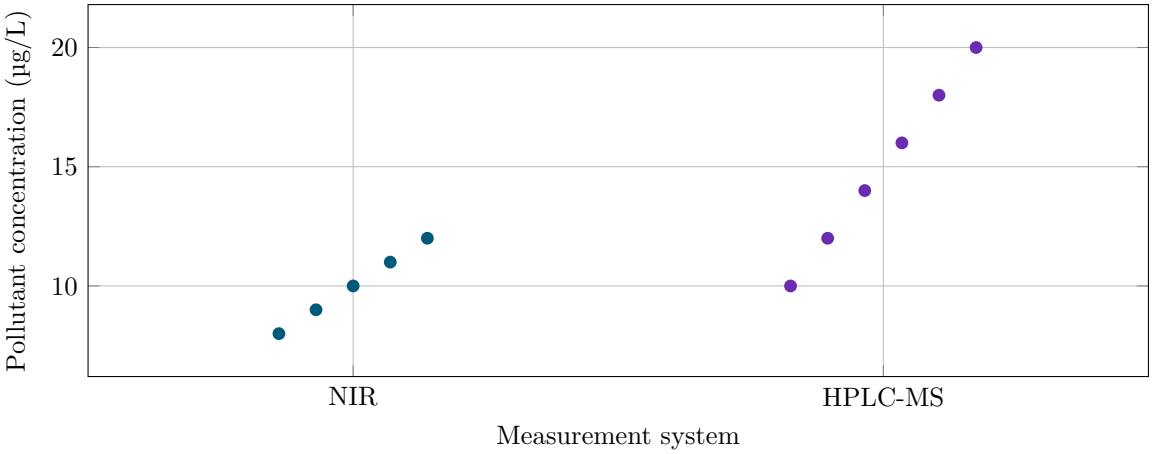


Table 3. Welch R-style report, approximate ν, two-sided p and mean difference.

Field	Result
Hypotheses	H_0: equal population means; H_1: different
Statistic / ν / p	$t = -2.9704$; $\nu \approx 6.972$; two-sided $p \approx 0.02089$
Mean estimates	NIR 10 µg/L (n 5); HPLC-MS 15 µg/L (n 6)
Mean difference	NIR minus HPLC-MS: −5 µg/L
Decision / model	Reject equal population means at 5%; independent Normal groups, unequal variances permitted. Accuracy needs separate reference evidence.

Fig. 21. NIST: independent and paired t comparisons.

NIST: paired, pooled and Welch t-tests

`https://www.itl.nist.gov/div898/handbook/eda/section3/eda353.htm`

5 Prespecification and a three-claim test-family

Good scientific practice starts with a population hypothesis, then plans the experiment and analysis before outcomes are known. Check assumptions, apply the prespecified procedure and report the result. Searching endpoints, directions, cutoffs or exclusions until p is small changes the error behaviour. Exploratory analyses can be useful when labelled honestly; they are not replacements for a confirmatory plan.

Consider independent replicate preparations from one water sample and three separate below-limit claims: chloride, bromate and phosphate, each against its own illustrative limit. Each H_0 is at or above the limit and H_1 below. The test-family error criterion here is at least one false below-limit claim. If all three nulls are true and the tests are independent, the chance of no false rejection is 0.95 cubed, or 0.857375. Thus at least one false rejection occurs with probability 0.142625, about 14.26%. Shared sample preparation can make the tests dependent, so that exact rate is an example, not a universal answer. This is not the same decision problem as one joint claim that all analytes pass; a conjunctive pass-only rule has different error logic.

Bonferroni in Eq. 5 divides the chosen 5% test-family level by three, or equivalently multiplies valid raw p-values by three and caps at one. Use only one route. Its union bound permits dependence. The illustrative table Table 4 has raw values **0.008, 0.021, 0.200**, adjusted to **0.024, 0.063, 0.600**. Only chloride supports a below-limit claim on this rule. Holm is another prespecified family-wise procedure: sort p, use successive thresholds 0.05/3, 0.05/2 and 0.05, and stop at the first failure. Here its adjusted values are 0.024, 0.042 and 0.200. Choose the procedure in advance. The experiment schematic Fig. 22 makes the three replicate series explicit.

Eq. 5. Bonferroni controls at least one false rejection in a prespecified test-family.

$$\alpha_j = \frac{\alpha_{\text{family}}}{m}, \qquad p_{j,\text{adj}} = \min(1, m p_j)$$

Symbols

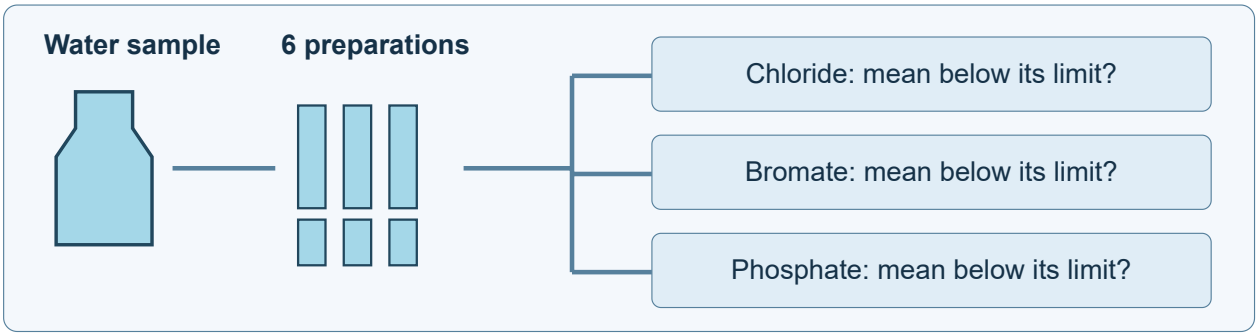
- α**_family**: Chosen test-family significance level, here 0.05.
- **m**: Number of planned tests, here three; positive integer.
- **j**: One test's index.
- α**_j**: Per-test significance level.
- **p_j; p_j,adj**: Valid raw p-value; adjusted p-value.
- **min**: Choose the smaller of the two arguments.
- **mp_j**: Multiplication of m and the raw p-value.
- **/, =, parentheses**: Divide, equal, group arguments.

All levels and probabilities are dimensionless. Test-family membership is prespecified. The union-bound control permits dependent tests; it controls at least one false rejection, not every possible joint decision rule.

Table 4. Three illustrative below-limit claims and their adjusted p-values.

Endpoint	Raw p	Bonferroni adjusted p
Chloride below its limit	0.008	0.024: reject at family 0.05
Bromate below its limit	0.021	0.063: do not reject
Phosphate below its limit	0.200	0.600: do not reject

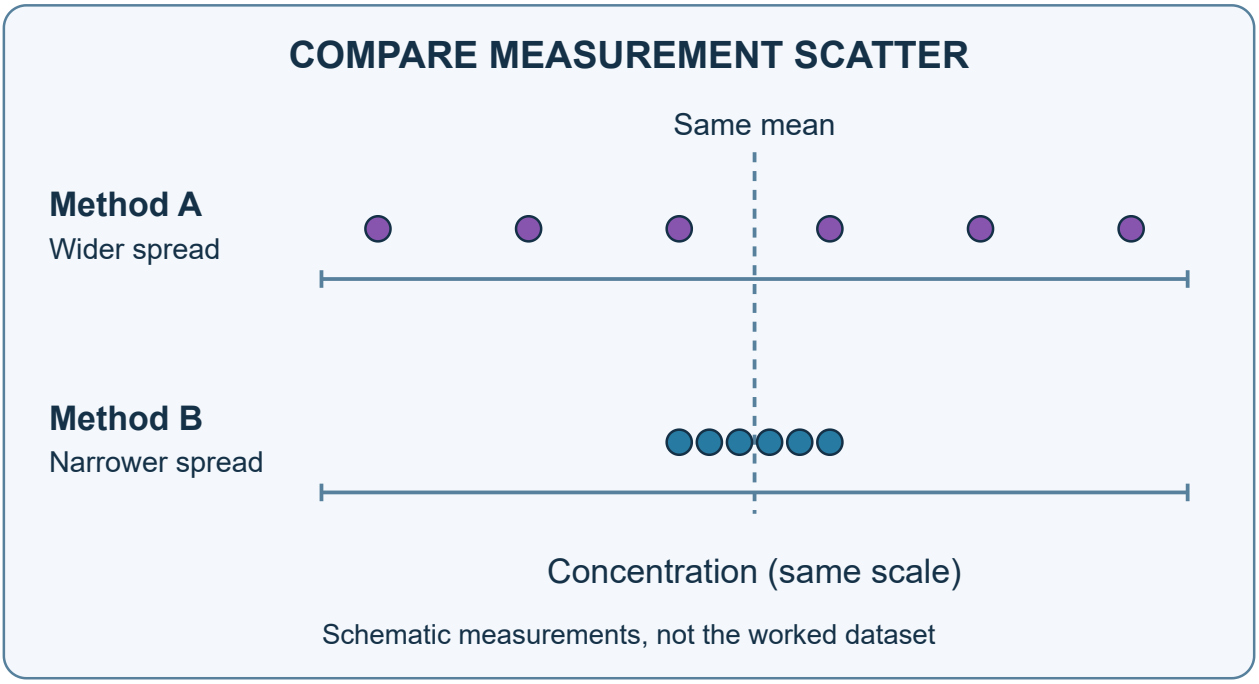
Fig. 22. One water sample, six preparations and three analytes.



6 F references for variances and nested calibration fits

The F section marker Fig. 23 opens comparisons of variances. As developed in Level 5, a ratio of independent Normal-sample variances has an F reference when population variances are equal. The **test statistic** Eq. 6 keeps ν_1 **and** ν_2 in order. In the precision study Fig. 24, SDs **0.20 and 0.10 mg/L** yield a variance ratio **four**. Six independent preparations per method give $(\nu_1, \nu_2) = (5, 5)$. The NIST chapter Fig. 25 explains this reference. The report Table 5 gives **upper-tail p = 0.07719**: do not reject the equality boundary against greater A variance. Non-rejection cannot prove equal precision. Normality is especially important for this exact reference.

Fig. 23. Variance-ratio theme of the F-test section.



Eq. 6. Prespecified variance-ratio orientation and ordered ν_1, ν_2.

$$F = \frac{s_1^2}{s_2^2}, \qquad (\nu_1, \nu_2) = (n_1 - 1, n_2 - 1)$$

Symbols

- **F**: Observed variance ratio; dimensionless.
- $\mathbf{s_1}^2$, $\mathbf{s_2}^2$: Sample variances, denominators $n_1 - 1$ and $n_2 - 1$; $(\text{mg/L})^2$.
- **1, 2**: Prespecified numerator A and denominator B.
- $\mathbf{n_1}$, $\mathbf{n_2}$: Independent sample sizes, each greater than one.
- ν_1, ν_2: Numerator and denominator degrees of freedom, in that order.
- **2; −; =; fraction bar**: Square SD, subtract, equal, divide.
- **Parentheses, comma**: Ordered pair of degrees of freedom.

Exact reference: independent Normal samples and equal positive population variances under H_0. The shown alternative is greater A variance, upper tail. A two-sided question needs both tails. Prespecify the ratio orientation.

Fig. 24. Six independent preparations per method and SD versus variance ratios.

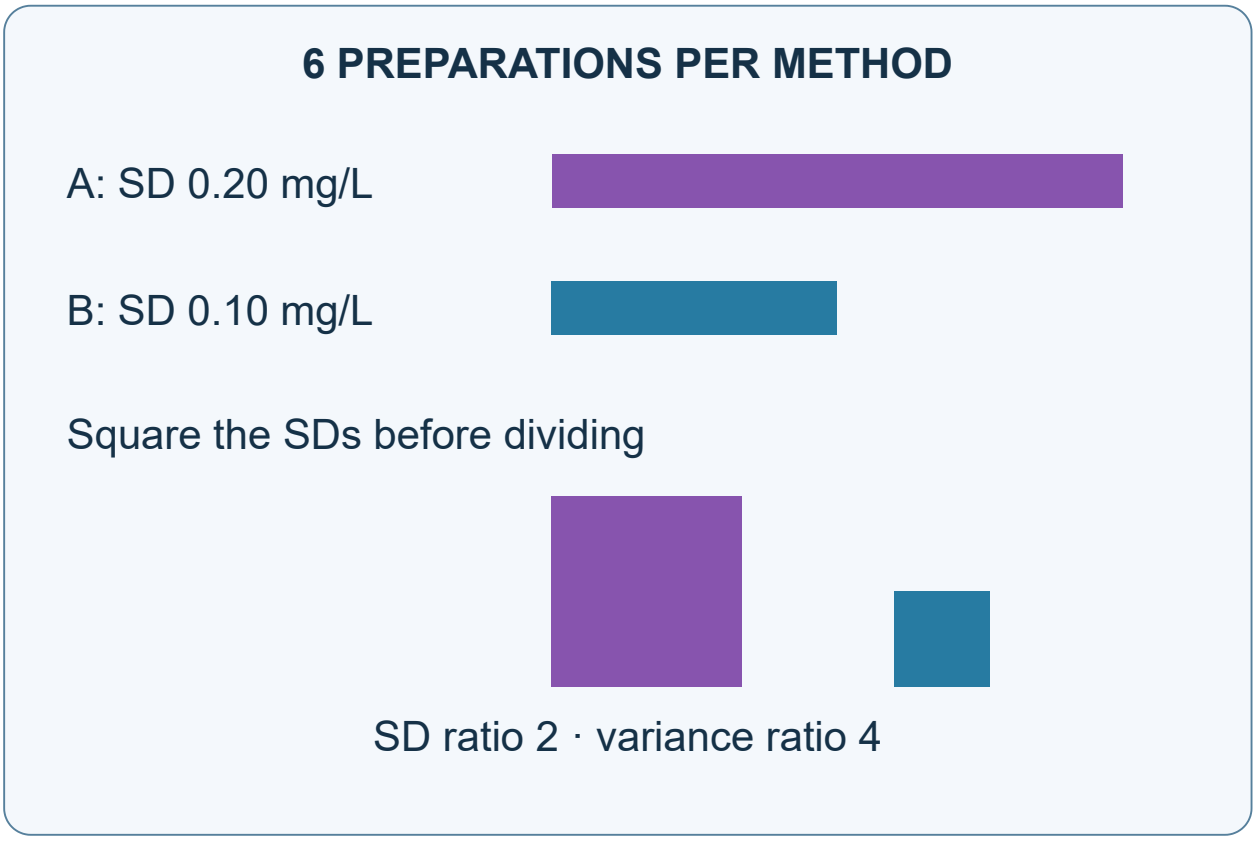


Fig. 25. NIST reference for comparing variances with F.

NIST: comparison of variances

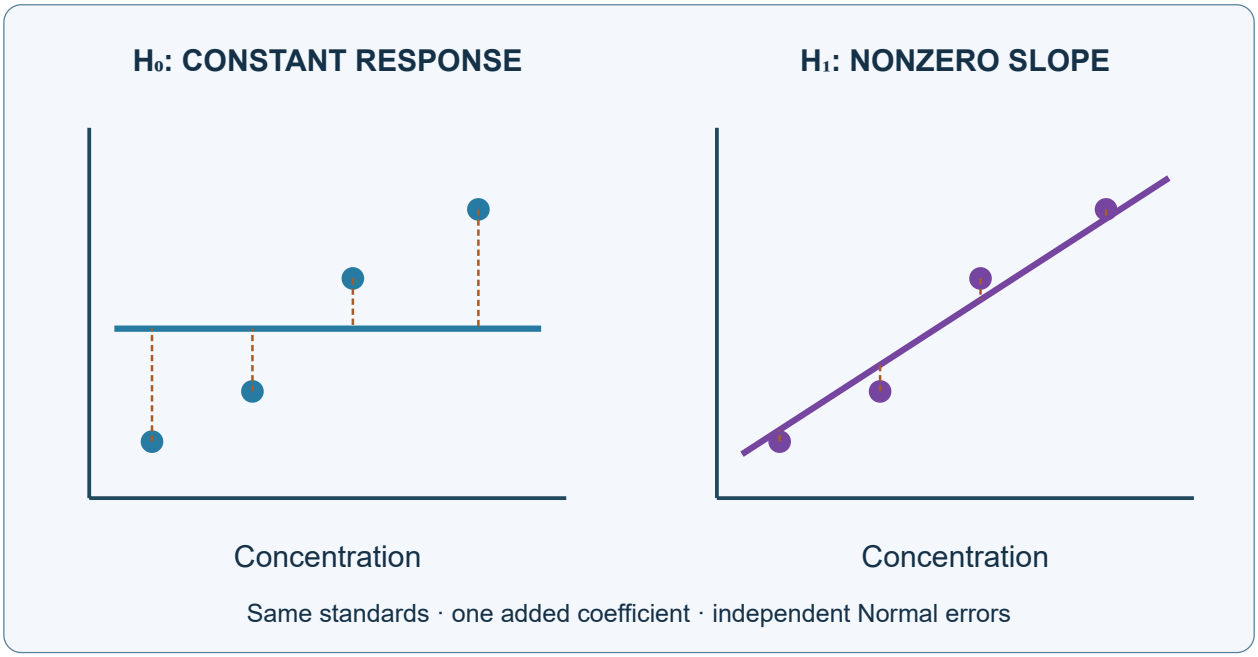
`https://www.itl.nist.gov/div898/handbook/eda/section3/eda359.htm`

Table 5. Upper-tail variance-comparison report, p 0.07719.

Field	Result
Hypotheses	H_0: equal population variances; H_1: A greater than B
F / (ν_1, ν_2) / p	$F = 4$; $(\nu_1, \nu_2) = (5, 5)$; upper-tail $p = 0.07719$
Estimates	Variance A 0.040, B 0.010 $(mg/L)^2$; ratio 4
Decision / model	Do not reject at 5%. Independent Normal samples; no proof of equal precision.

An F-test can instead compare nested models fitted to the same observations. The schematic calibration Fig. 26 contrasts a constant response with adding a concentration slope. Under the shared independent Normal-error model, one added coefficient gives F equal to the square of its corresponding coefficient t statistic. This F question detects either slope direction. It does not require two separate datasets or two unrelated raw variance estimates.

Fig. 26. Flat model versus nonzero slope using the same standards; schematic geometry.



Adding curvature to a straight line is another nested question. The formula Eq. 7 compares error reduction per added coefficient with remaining full-model error per residual degree of freedom. The authored summary in Fig. 27 uses ten standards, two versus three fitted coefficients, and RSS **12 versus 5**; the curves are conceptual. The report Table 6 gives $\mathbf{F = 9.8}$, $\mathbf{(\nu_1,\ \nu_2) = (1,\ 7)}$, $\mathbf{p = 0.01660}$. Reject the zero-curvature coefficient under the shared Normal-error model. Residual checks and validation of the working range remain necessary. NIST describes related conditional coefficient tests in Fig. 28; for one added coefficient, the nested F equals its squared t statistic.

Eq. 7. Nested-model statistic from additional explained sum of squares and remaining error.

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_F)/(p_F - p_R)}{\mathrm{RSS}_F/(n - p_F)}$$
$$(\nu_1, \nu_2) = (p_F - p_R, n - p_F)$$

Symbols

- **F**: Observed dimensionless test statistic.
- **R, F subscripts**: Reduced and full model.
- **RSS_R, RSS_F**: Residual sums of squares; squared response unit.
- **p_R, p_F**: Fitted independent coefficients, including intercept; full has more.
- **n**: Same observations in both models; exceeds full coefficient count.
- ν_1, ν_2: Numerator and denominator degrees of freedom.
- **−, =, fraction bars**: Subtract, equal, divide.
- **Parentheses, comma**: Grouping and ordered degrees of freedom.

Exact reference: full-rank nested linear models; independent Normal errors with common positive variance. H_0 sets added coefficients to zero. More flexible fits can explain random scatter; the F comparison calibrates the improvement.

Fig. 27. Authored line/curve RSS summaries 12 and 5 for ten standards.

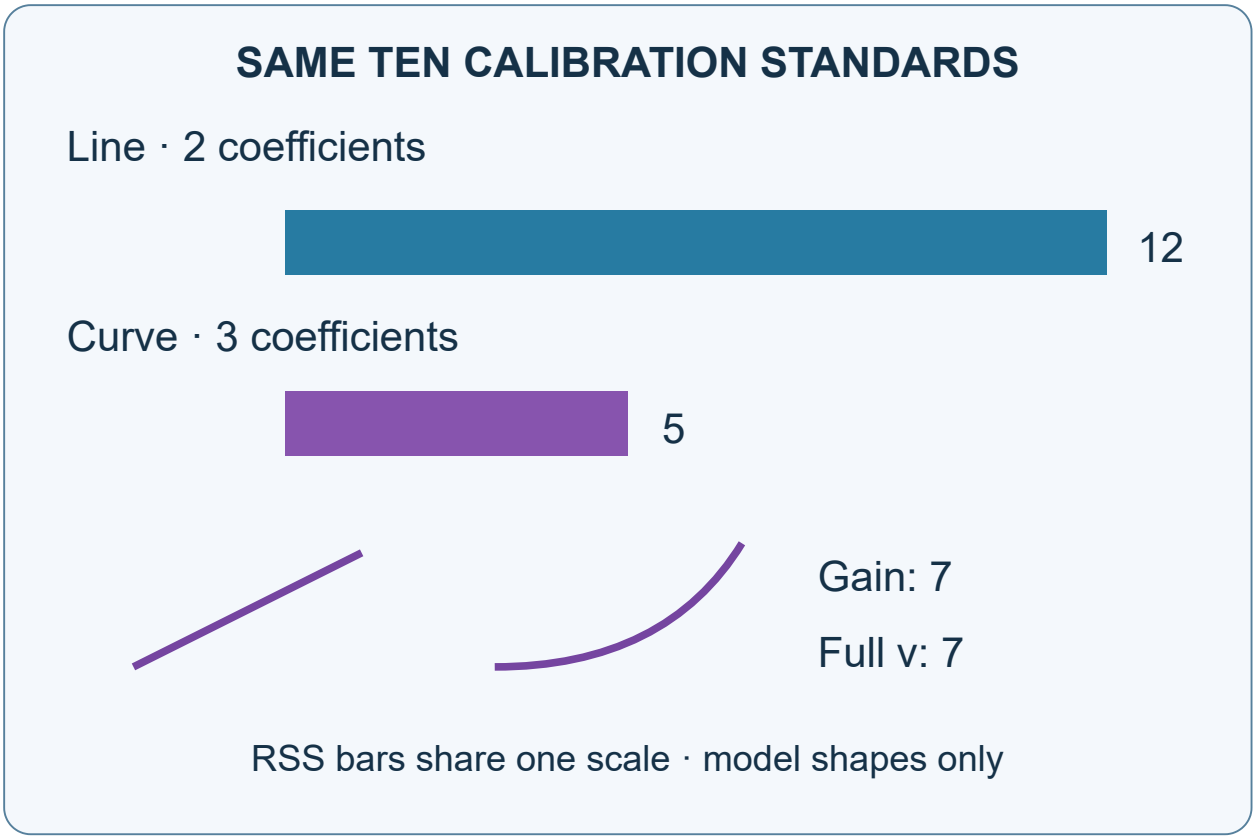


Table 6. Nested curvature-test report, with upper-tail p and model assumptions.

Field	Result
Hypotheses	H_0: curve coefficient zero; H_1: nonzero
F / (ν_1, ν_2) / p	F = 9.8; $(\nu_1, \nu_2) = (1, 7)$; upper-tail p = 0.01660
Same ten standards	Line: 2 coefficients, RSS 12; curve: 3 coefficients, RSS 5
Decision / model	Reject at 5%; independent Normal errors, common variance, nested full-rank fits. Inspect residuals and working range too.

Fig. 28. NIST: conditional regression coefficient tests.

NIST: regression coefficient tests

`https://www.itl.nist.gov/div898/handbook/pmd/section4/pmd447.htm`

7 Three distinct chi-square questions

The overview Fig. 29 separates a numeric variance question from two questions about counts. For a fixed known target variance, the setup Fig. 30 leads to Eq. 8, an exact Normal-model reference. The six results in Fig. 31 have sample variance $\mathbf{0.024\ (mg/L)^2}$. A known SD target $\mathbf{0.10\ mg/L}$ gives variance $\mathbf{0.010\ (mg/L)^2}$. Multiply by $\nu = \mathbf{5}$ and divide by the target: $\mathbf{Q = 12}$. The report Table 7 gives **upper-tail p = 0.03479**, evidence of excess variance at 5%. A separately estimated target requires an analysis including its uncertainty.

Fig. 29. Measurement scatter, particle composition and classification association are different questions.

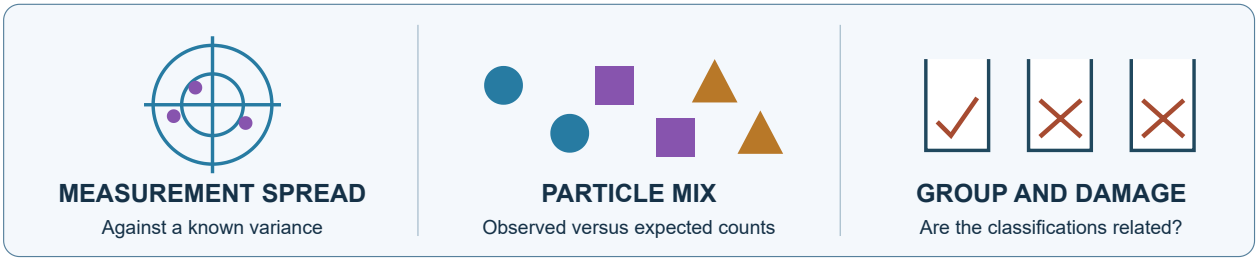
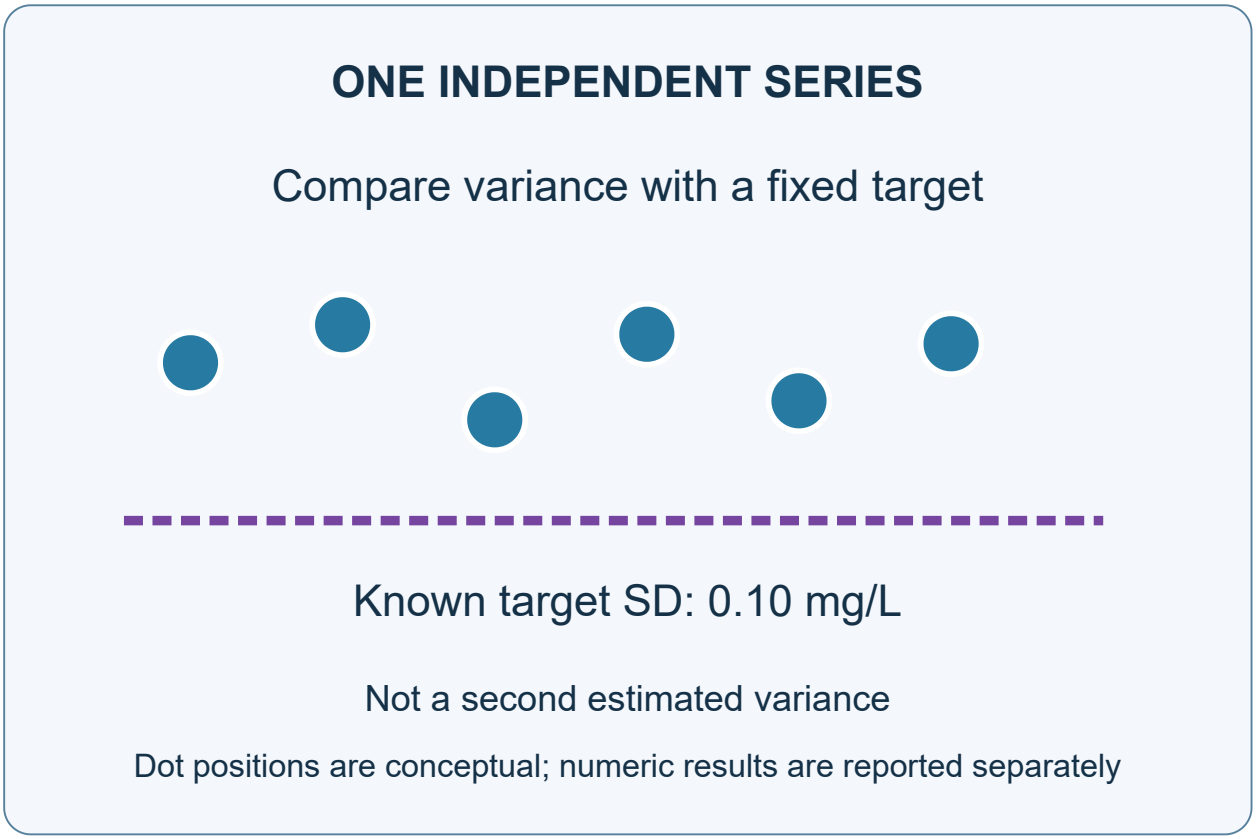


Fig. 30. One independently measured series against a known variance target.



Eq. 8. Exact known-variance Normal-model statistic.

$$Q = \frac{(n-1)s^2}{\sigma_0^2}, \qquad \nu = n - 1$$

Symbols

- **Q**: Observed chi-square statistic; dimensionless.
- **n**: Number of independent measurements, greater than one.
- ν: Residual degrees of freedom.
- $\mathbf{s}^2$: Sample variance, denominator n − 1; $(\mathrm{mg/L})^2$.
- ${\sigma_0}^2$: Known positive target population variance; $(\mathrm{mg/L})^2$.
- **0; 2**: Null-reference subscript; exponent squares SD.
- **−, =, fraction bar**: Subtract, equal, divide.
- **Juxtaposition; parentheses**: Multiply; group n − 1.

Exact reference assumes independent identically distributed Normal measurements and a fixed target variance. For an at-or-below null, use equality as the upper-tail boundary. An estimated reference variance instead needs a two-sample model.

Fig. 31. Six synthetic measurements with observed variance 0.024 $(\mathrm{mg/L})^2$.

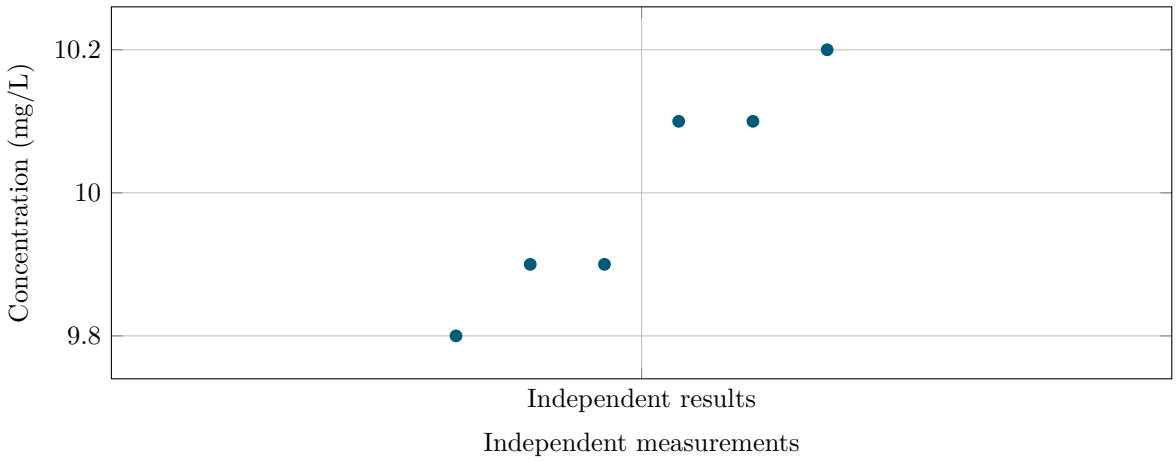


Table 7. One-variance report with upper-tail p.

Field	Result
Hypotheses	H_0: variance at or below 0.010 $(\mathrm{mg/L})^2$; H_1: above
Statistic / ν / p	Q = 12; ν = 5; upper-tail p = 0.03479
Sample / reference	n = 6; variance 0.024 $(\mathrm{mg/L})^2$; known target SD 0.10 mg/L
Decision / model	Reject at 5%; independent Normal observations, known fixed target. Evidence of excess scatter.

For particle categories, Pearson's **test statistic** Eq. 9 adds squared observed-minus-expected count differences divided by expected count. The bars Fig. 32 retain **60 independent particles**: observed

30,20,10, expected **30,18,12** from fixed shares **50%,30%,20%**. The report Table 8 gives $\mathbf{X^2 = 0.5556}$, $\boldsymbol{\nu = 2}$, **approximate upper-tail p = 0.7575**. Do not reject the mix; this cannot prove its shares. Categories must be exhaustive and mutually exclusive. Fitted shares or sparse expectations require suitable calibration.

Eq. 9. Pearson category-count discrepancy and fixed-probability ν.

$$X^2 = \sum_{j=1}^{k} \frac{(O_j - E_j)^2}{E_j}$$
$$E_j = N\pi_j, \qquad \nu = k - 1$$

Symbols

- $\mathbf{X^2}$: Dimensionless Pearson statistic.
- **j; k**: Category index; number of exhaustive, mutually exclusive categories.
- **O_j; E_j**: Observed count; positive null expected count.
- **N**: Total independent particles.
- π**_j**: Fixed category probability; all probabilities add to one.
- ν: Approximate reference degrees of freedom with fixed probabilities.
- Σ: Sum over categories 1 through k.
- **2, −, =, fraction bars**: Square, subtract, equal, divide.

Counts, probabilities and statistic are dimensionless. Juxtaposition multiplies; parentheses group. Adequate expected counts support the chi-square approximation. Fitted probabilities require adjusted degrees of freedom and calibration.

Fig. 32. Observed versus predicted category counts on a shared scale.

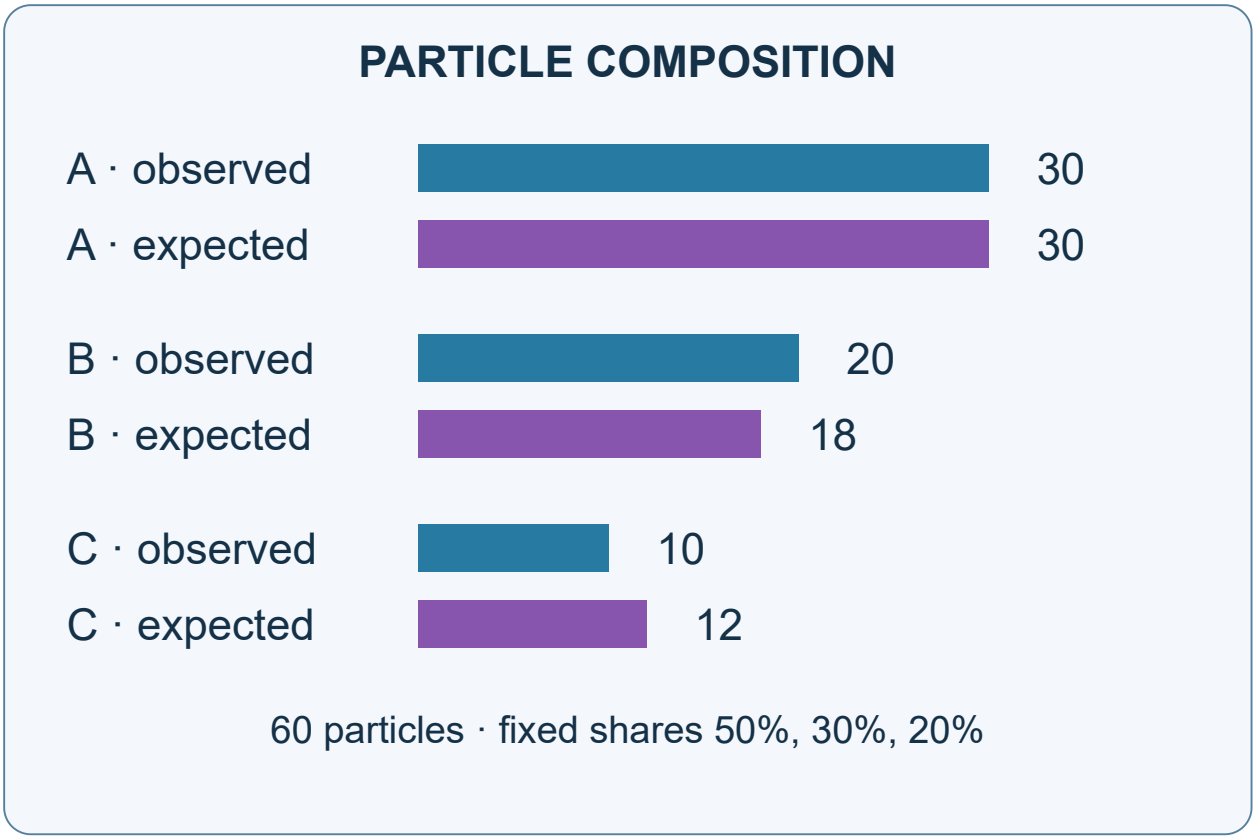


Table 8. Goodness-of-fit report with approximate chi-square reference.

Field	Result
Hypotheses	H_0: proportions 50%, 30%, 20%; H_1: a different mix
Statistic / ν / p	Pearson $X^2 = 0.5556$; $\nu = 2$; upper-tail approximate p = 0.7575
Observed / expected	30, 20, 10 / 30, 18, 12; 60 independent particles
Decision / model	Do not reject at 5%. Fixed multinomial probabilities, adequate expected counts. No proof of the mix.

For group/damage independence, Eq. 10 obtains expected cells from row and column totals. The synthetic table Fig. 33 has control **8 affected/12 unaffected** and exposure **14/6**. Independence predicts **11 affected and 9 unaffected** in each group of twenty. The report Table 9 uses uncorrected Pearson: $\mathbf{X^2 = 3.6364}$**,** ν **= 1, approximate upper-tail p = 0.05653**. Do not reject at 5%; this cannot prove independence. Every independent vessel enters one cell. The setup Fig. 34 maps treatment to rows and classification to columns. R's default continuity correction gives a different statistic. Further reading: NIST variance Fig. 35, goodness of fit Fig. 36, and contingency tables Fig. 37.

Eq. 10. Independence-predicted cell counts and reference ν.

$$X^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$
$$E_{ij} = \frac{R_i C_j}{N}$$
$$\nu = (r-1)(c-1)$$

Symbols

- **X^2**: Dimensionless Pearson discrepancy statistic.
- **i, j; r, c**: Row/column indices; numbers of categories, each greater than one.
- **O_ij; E_ij**: Observed cell count; positive expected count under independence.
- **R_i, C_j**: Row total and column total.
- **N**: Positive grand total of independent vessels.
- ν: Approximate reference degrees of freedom.
- **Double** Σ: Sum over every row and column cell.
- **2; juxtaposition; fraction bars**: Square, multiply, divide.
- **$-$, $=$, parentheses**: Subtract, equal, group.

Counts, degrees of freedom and statistic are dimensionless. Each independent vessel enters one cell. Adequate expected counts support the chi-square approximation; sparse cells need suitable exact or simulation calibration. The vessel is the independent unit.

Fig. 33. Affected/unaffected vessel counts and their independence prediction.

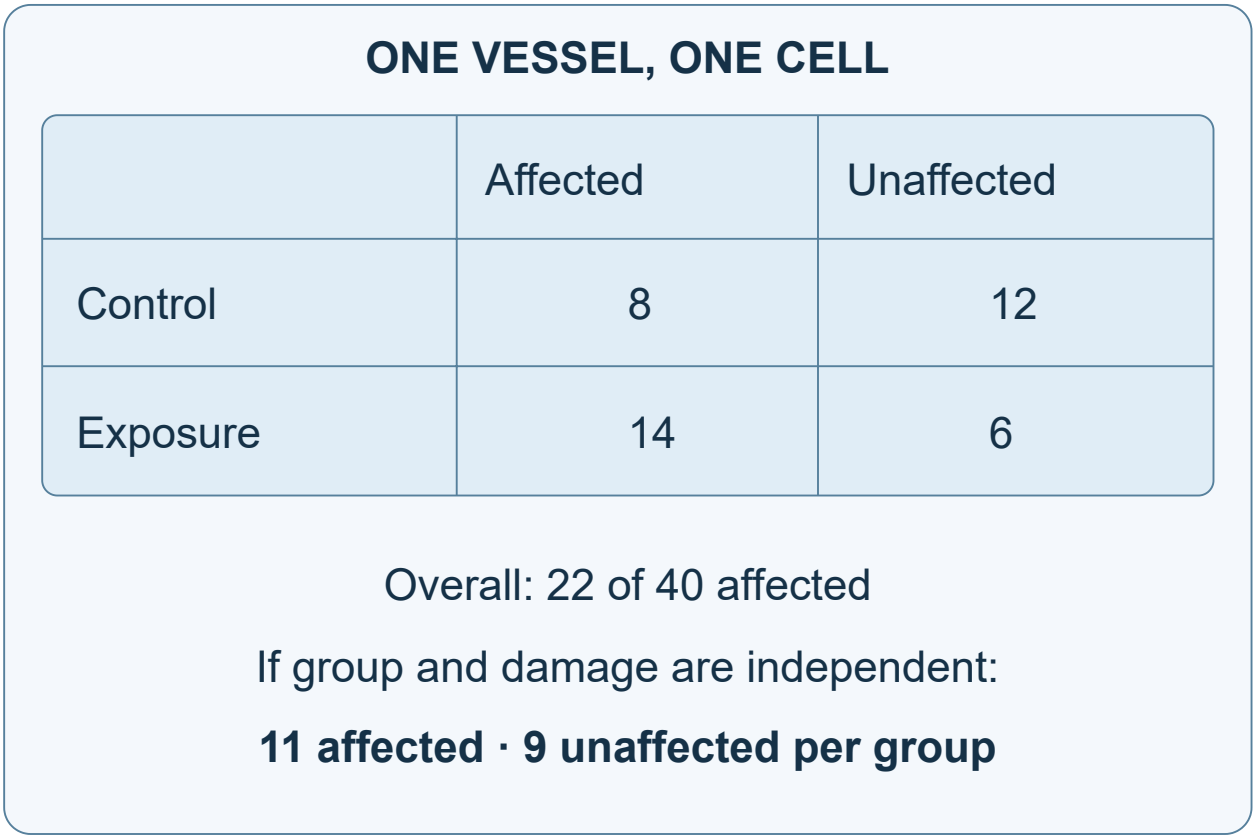


Table 9. Uncorrected Pearson independence report, approximate p 0.05653.

Field	Result
Hypotheses	H_0: group and damage independent; H_1: associated
Statistic / ν / p	$X^2 = 3.6364$; $\nu = 1$; upper-tail approximate p = 0.05653
Convention / units	Uncorrected Pearson; 40 independent vessels; expected 11 affected / 9 unaffected per group
Decision at 5%	Do not reject. Adequate expected counts support the approximation; not evidence of no association.

Fig. 34. How the vessels become table rows and columns.

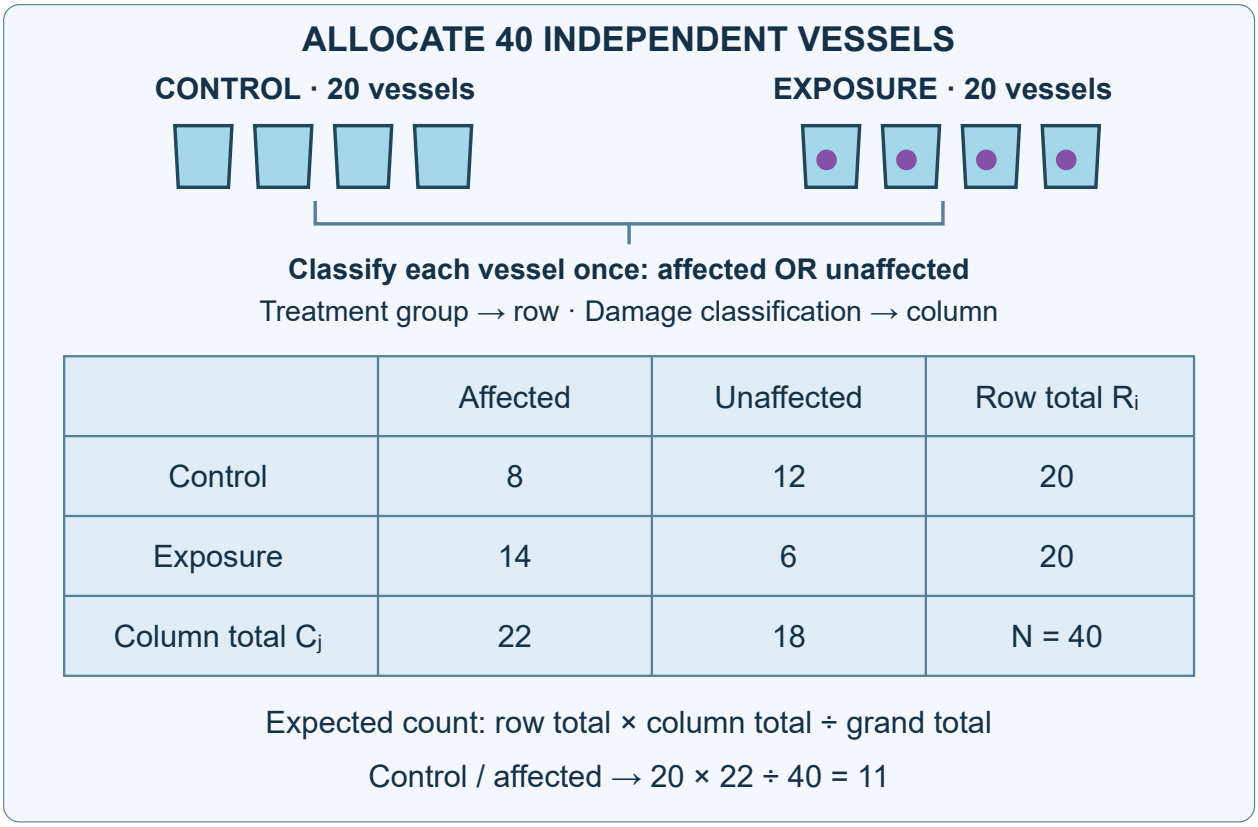


Fig. 35. NIST: known-variance test.

NIST: known-variance chi-square test

`https://www.itl.nist.gov/div898/handbook/eda/section3/eda358.htm`

Fig. 36. NIST: goodness-of-fit reference.

NIST: chi-square goodness of fit

`https://www.itl.nist.gov/div898/handbook/eda/section3/eda35f.htm`

Fig. 37. NIST: contingency-table independence.

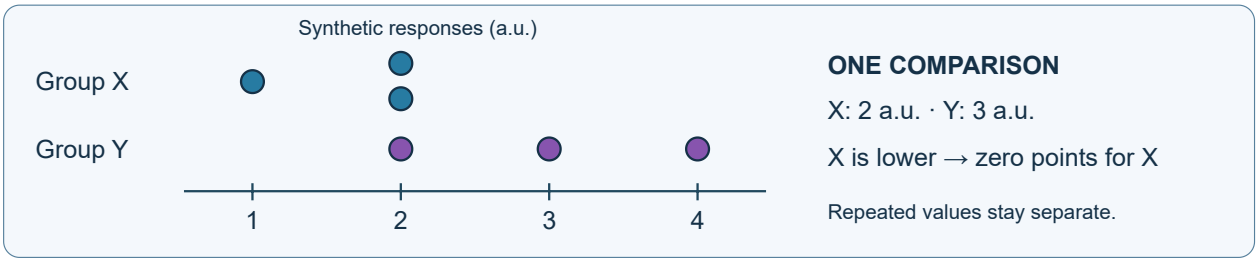
NIST: contingency-table independence

`https://www.itl.nist.gov/div898/handbook/prc/section4/prc45.htm`

8 Independent ordering and original paired differences

Rank methods use order rather than raw mean distance, but do not remove the need to specify sampling and a null. The independent groups in Fig. 38 contain X values 1,2,2 and Y values 2,3,4. The U statistic Eq. 11 scores an X win one, a tie one half and a loss zero. The grid Fig. 39 gives zero wins, seven losses and two ties: U for X is one. The usual null makes group labels exchangeable under equal distributions. A median-shift interpretation additionally needs the same-shaped location-shift model; differing shapes and detection-limit censoring need care.

Fig. 38. Independent numeric groups on a shared ordered response scale.



Eq. 11. Oriented Mann–Whitney statistic with half credit for ties.

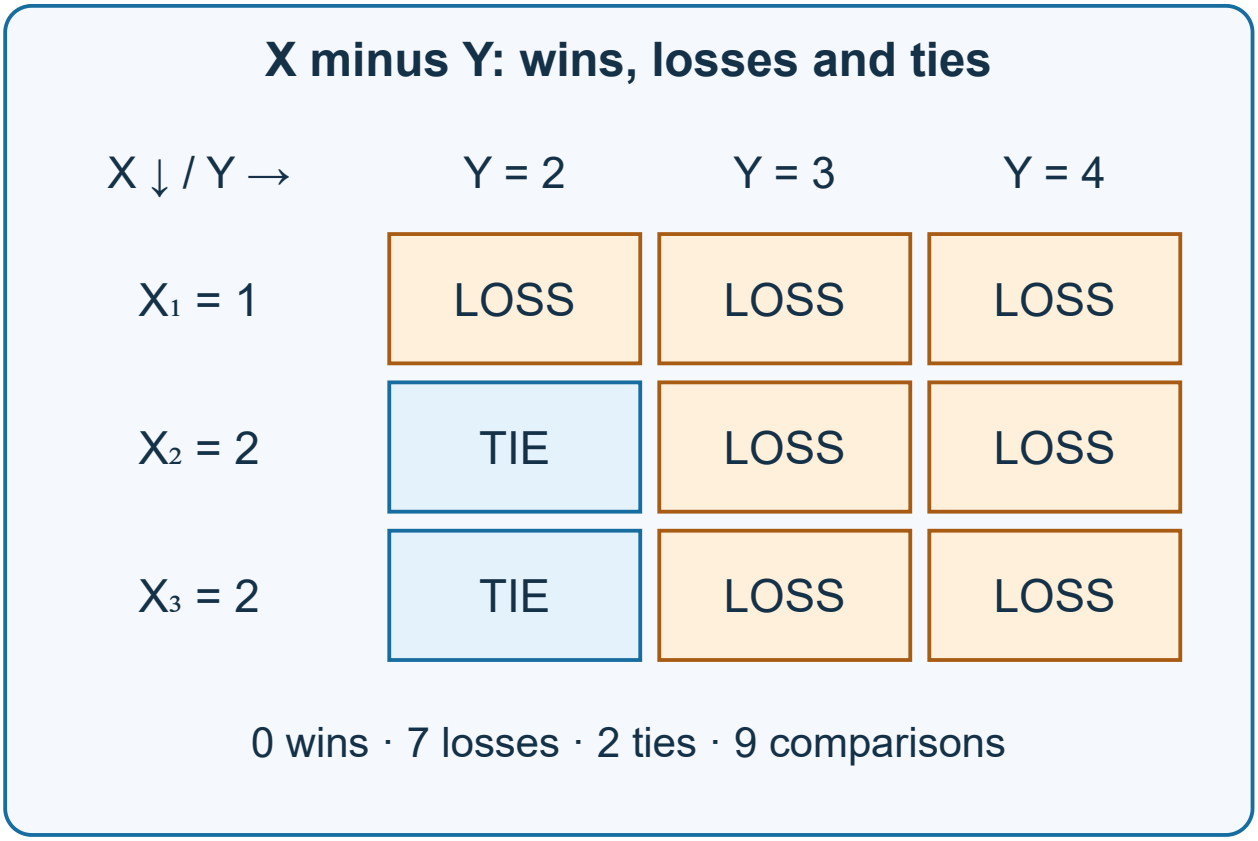
$$U_X = \sum_{i=1}^{n_X} \sum_{j=1}^{n_Y} \left[\mathbf{1}(x_i > y_j) + \tfrac{1}{2}\mathbf{1}(x_i = y_j) \right]$$

Symbols

- **X, Y**: Independent groups on one ordered response scale.
- **n_X, n_Y**: Positive group sample sizes.
- **i, j; x_i, y_j**: Observation indices and their measured values.
- **U_X**: X wins plus half the ties; dimensionless.
- **Bold 1(event)**: One when the condition holds, otherwise zero.
- $>$, $=$, $+$, $\frac{1}{2}$: Greater than, equal, add, half credit.
- **Double** Σ: Visit every X–Y comparison.
- **Brackets; juxtaposition**: Group terms; multiply.

Usual H_0: identical group distributions, exchangeable labels. This example uses an explicitly tie-aware label-permutation reference. Median-shift interpretation additionally needs common distribution shape. Software may report the complementary U.

Fig. 39. All nine cross-group comparisons; repeated observations are not merged.



The explicit tie-aware reference in Table 10 enumerates all twenty assignments of three of the six observations to X, preserving tied observations as distinct units. Six assignments give U at least as far from its null centre 4.5 as the observed value one. Two-sided p is therefore 0.300, and we do not reject at 5%. This convention is declared rather than presenting a no-tie exact table as applicable. The observed tendency for X to be smaller is descriptive, not by itself a significant distributional result.

Table 10. Exact tie-aware two-sided label-permutation result and interpretation.

Field	Result
Null / statistic	Equal distributions; U for X = 1; null centre 4.5
Exact two-sided p	0.300; 6 of 20 label assignments at least as far from 4.5. Tied observations retained.
Interpretation	X tends lower in the sample; do not reject at 5%. Independent exchangeable labels under H_0; median-shift interpretation needs common shape.

For paired observations, keep the original pairs. Form one **difference** per pair, then calculate the **rank of abs(diff)** using Eq. 12. Here abs means absolute value. In Fig. 40, differences **1,−2,3,4,5 µg/L** give $\mathbf{W^+ = 13}$ and negative-rank sum **2**. Under the symmetric zero-shift H_0, each independent difference has two equally likely signs. Five choices give $\mathbf{2 \times 2 \times 2 \times 2 \times 2 = 32}$ **patterns**. The total rank sum is **15** and the null centre **7.5**. Three patterns give $\mathbf{W^+ \leq 2}$,

and three give $\mathbf{W^+ \geq 13}$: $\mathbf{p = 6/32 = 0.1875}$, as reported in Table 11. At 5%, do not reject H_0. State zero and tie conventions in software.

Eq. 12. Signed ranks of original within-pair differences, not independently sorted samples.

$$d_i = x_i - y_i, \qquad W^+ = \sum_{i:d_i>0} \operatorname{rank}(|d_i|)$$

Symbols

- **i**: Original independent pair identity, not sorted-value position.
- **x_i, y_i; d_i**: Pair readings and their first-minus-second difference; same response unit.
- **|d_i|**: Absolute difference size.
- **rank**: Rank nonzero absolute differences; average ranks for ties.
- $\mathbf{W^+}$: Sum of ranks of positive differences; dimensionless. Superscript + labels this sum.
- **Σ; colon**: Sum over indices satisfying the condition; such that.
- **>, −, =**: Strictly positive comparison, subtract, equal.
- **Parentheses**: Group the rank argument.

Usual shift inference requires independent symmetric differences. Remove zeros before ranking in this convention. Do not independently sort X and Y. Declare tie/zero handling and exact versus approximate calculation.

Fig. 40. Five independent paired differences become signed size ranks.

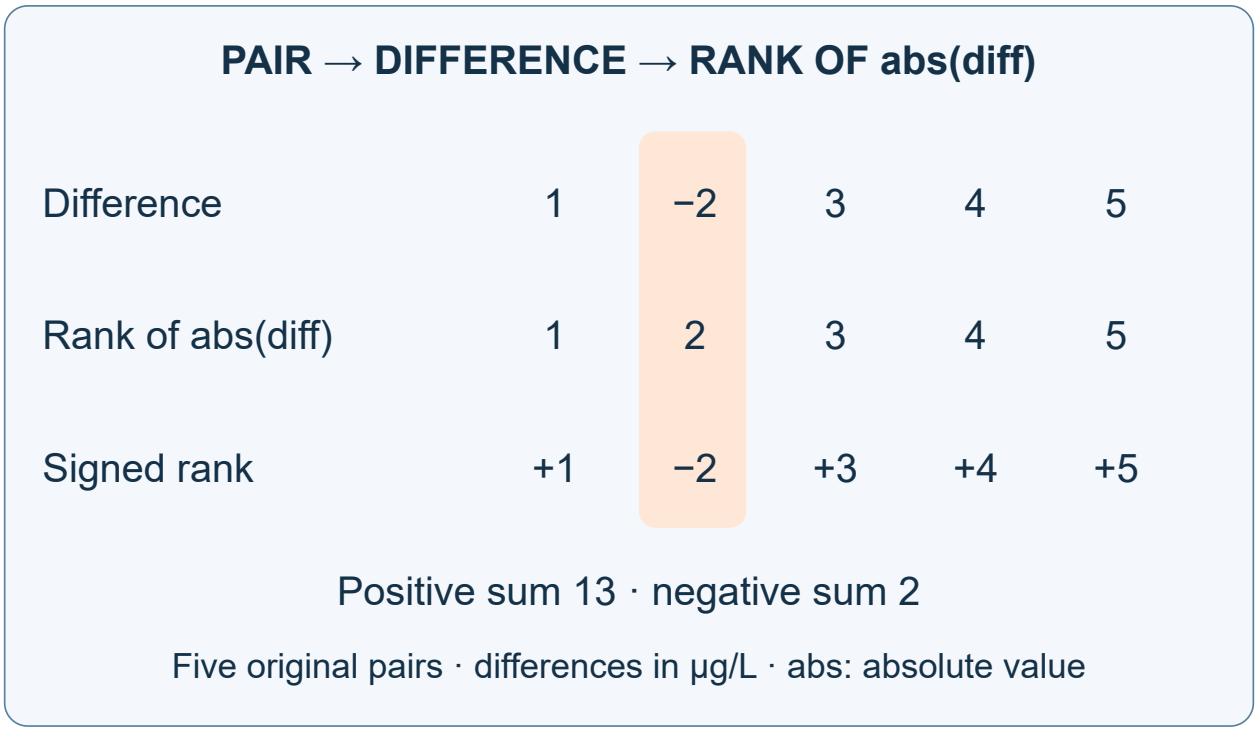


Table 11. Exact signed-rank report with symmetric-difference assumptions.

Field	Result
Null / statistic	Symmetric zero-shift differences; positive-rank sum 13, negative sum 2
Exact two-sided p	0.1875; 6 of 32 sign assignments; no zeros or absolute ties
Decision / units	Do not reject at 5%. Five independent paired changes in µg/L; symmetry required for the usual shift inference. Do not sort the two raw samples separately.

Try the reusable compare–score–accumulate interaction Fig. 41 before reading the grid. Each click brings one original X/Y pair together and credits the X account. The NIST rank-sum Fig. 42 and signed-rank Fig. 43 pages provide further reading; software statistic orientation and tie conventions can differ. Our declared tie-aware label-permutation calculation remains the reference for this U example.

Fig. 41. Compare one pair, credit its points, accumulate U.

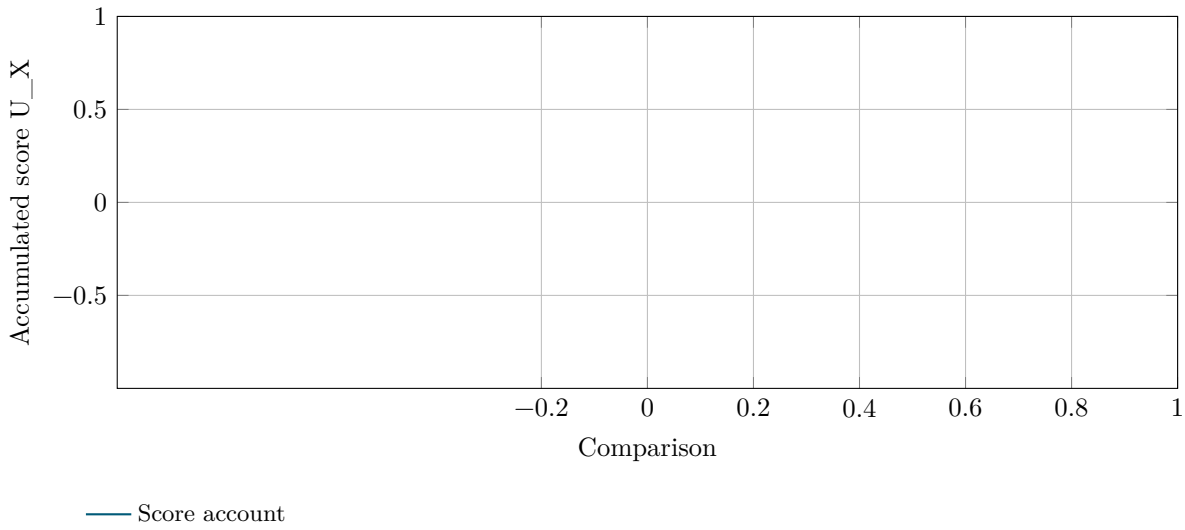


Fig. 42. NIST rank sum: check statistic orientation and tie conventions.

NIST: rank-sum test (software conventions differ)

`https://www.itl.nist.gov/div898/software/dataplot/refman1/auxillar/ranksum.htm`

Fig. 43. NIST: signed-rank test.

NIST: signed-rank test

`https://www.itl.nist.gov/div898/software/dataplot/refman1/auxillar/signrank.htm`

9 Effect magnitudes and their interpretation

A **p-value** assesses compatibility with the null model. **Effect magnitude** describes the size of the observed difference. Start with the original units: NIR minus HPLC-MS is **−5 µg/L**. The sampling design limits the scope of inference.

Cohen's d in Eq. 13 divides the independent-group mean difference by pooled within-group descriptive SD. Here **d = −1.677**: NIR reads lower and the mean difference is about **1.68 pooled SDs**. The reference table Table 12 classifies absolute d: below **0.2 negligible**, **0.2 to below 0.5 small**, **0.5 to below 0.8 medium**, and **at least 0.8 large**. Our result is **large** by this convention. These effsize-manual bands describe a standardised magnitude; chemical relevance depends on the application. The mean inference remains Welch.

Eq. 13. Native-unit mean difference and descriptive pooled-SD Cohen's d.

$$\widehat{\Delta} = \bar{x} - \bar{y}, \qquad d = \frac{\bar{x} - \bar{y}}{s_p}$$

Symbols

- $\widehat{\Delta}$: Estimated X-minus-Y mean difference; hat marks estimation.
- $\bar{x}$, $\bar{y}$: Independent-group means; here NIR and HPLC-MS in µg/L.
- s_p: Pooled within-group sample SD; positive square root of variance weighted by residual degrees of freedom; µg/L.
- **d**: Uncorrected dimensionless Cohen effect estimate.
- **p subscript**: Labels pooling.
- **−, =, fraction bar**: Subtract, equal, divide.

Here pooling defines a descriptive standardisation. The mean test remains Welch. Magnitude guides are conventions; report native units, uncertainty and design. Paired effects use other denominators.

Table 12. Conventional bands for absolute Cohen d.

Absolute Cohen d	Conventional description
Below 0.2	Negligible
0.2 to below 0.5	Small
0.5 to below 0.8	Medium
0.8 or more	Large

Cliff's delta in Eq. 14 is the balance of cross-group wins and losses divided by every comparison, including ties. Our earlier grid gives −7/9, about −0.778, indicating X tends lower. Conventional absolute-delta bands are below 0.147 negligible, 0.147 to below 0.33 small, 0.33 to below 0.474 medium, and at least 0.474 large, following the Romano thresholds documented in effsize. They do not set significance or practical-relevance criteria. Comparisons reuse original observations, so uncertainty estimation must respect the independent units, not pretend there are nine independent observations.

Eq. 14. Cliff's delta from oriented wins, losses and the tie-aware U convention.

$$\hat{\delta} = \frac{N_{>} - N_{<}}{n_X n_Y} = \frac{2U_X}{n_X n_Y} - 1$$

Symbols

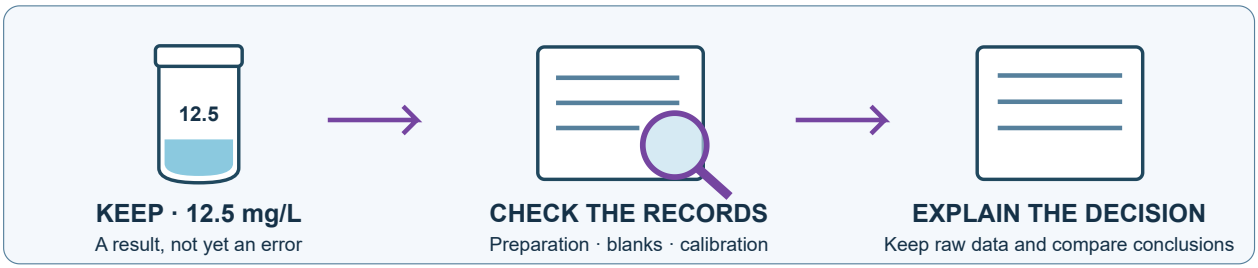
- $\hat{\delta}$: Estimated dimensionless Cliff effect from -1 to 1; hat marks estimation.
- **N_>, N_<**: Counts of X-over-Y wins and losses.
- **n_X, n_Y**: Positive independent-group sample sizes.
- **U_X**: X wins plus half the ties.
- **>, <**: Strictly greater and smaller comparisons.
- **2; −; =; fraction bars**: Twice, subtract, equal, divide.
- **Juxtaposition**: Multiply sample sizes.

Ties add zero to the signed balance but stay in the denominator. Positive delta means X tends larger. Pairs sharing observations are dependent; uncertainty follows the original independent units, not the number of comparisons.

10 A Grubbs flag starts an investigation

Five nitrate results are near ten mg/L and one is 12.5. Preparation errors, contamination, instrument behaviour and genuine pollution events can all produce unusual observations. The investigation picture Fig. 44 begins with keeping the raw value and its sample identity. Check acquisition order, blanks, calibration and preparation records rather than starting with deletion.

Fig. 44. Keep, investigate and explain the high nitrate result.



Grubbs' statistic Eq. 15 divides the largest absolute deviation from the sample mean by sample SD, calculated from all six original results. Its null is an independent Normal sample, with an alternative concerning one extreme. The two-sided critical construction Eq. 16 accounts for searching the complete sample. For 9.8,9.9,10.0,10.1,10.2,12.5 mg/L, G is about 2.0219 and the 5% cutoff about 1.8871. Compare observed statistic with critical value, just as other tests compare a statistic with a reference rejection region.

Eq. 15. Observed Grubbs statistic from the complete original sample.

$$G = \frac{\max_{1 \le i \le n} |x_i - \bar{x}|}{s}$$

Symbols

- **G**: Observed dimensionless maximum standardised deviation.
- **n; i**: Sample size, at least three; observation index.
- **x_i; $\bar{x}$; s**: Observation, sample mean, positive sample SD; all mg/L.
- **max; 1 ≤ i ≤ n**: Select the largest value over the complete sample.
- **| |**: Absolute magnitude; checks either extreme.
- **−, =, fraction bar**: Subtract, equal, divide.

Use all original values to compute mean and SD; sample SD uses denominator n − 1. H_0 is an independent Normal sample; the alternative concerns one extreme. The cutoff accounts for selecting the maximum.

Eq. 16. Selection-aware two-sided critical value and full symbol definitions.

$$c = t_{1-\alpha/(2n),n-2}$$
$$G_{\text{crit}} = \frac{n-1}{\sqrt{n}}\sqrt{\frac{c^2}{n-2+c^2}}$$

Symbols

- **n**: Number of observations, at least three.
- α: Prespecified two-sided significance level.
- **t_(q,ν)**: t quantile at cumulative probability q and degrees of freedom ν.
- **q; ν**: Here 1 minus alpha divided by twice n; n minus two degrees of freedom.
- **c**: Abbreviation for that dimensionless quantile.
- **G_crit**: Dimensionless Grubbs cutoff; crit labels the threshold.
- **2; $\sqrt{}$**: Squaring; positive square root.
- **+, −, =, fraction bars**: Add, subtract, equal, divide.

Juxtaposition multiplies; parentheses group. Reject if G exceeds G_crit under the independent Normal single-outlier model. This is a two-sided cutoff; one-sided procedures use a different allocation.

Since G exceeds its cutoff, flag **12.5 mg/L** under this model. A flag starts an investigation. Preserve the raw record and document any scientifically justified correction or exclusion. NIST explains the single-outlier reference in Fig. 45.

Fig. 45. NIST: single-outlier Grubbs test.

NIST: Grubbs single-outlier test

`https://www.itl.nist.gov/div898/handbook/eda/section3/eda35h1.htm`

For several suspected outliers, plan a **multiple-outlier procedure**. Generalized ESD Fig. 46 needs an upper bound **r** chosen before testing, an approximately Normal background, and its own adjusted sequential critical values. It calculates successive extreme-deviation statistics and selects the largest qualifying step. Repeated unadjusted Grubbs deletion changes error behaviour; a simple Bonferroni adjustment alone does not resolve adaptive deletion and masking. NIST reports reasonable critical-value approximation from $\mathbf{n} \geq \mathbf{15}$, stronger from $\mathbf{n} \geq \mathbf{25}$. The six-point example above remains a single-outlier illustration.

Fig. 46. NIST: Generalized ESD procedure and sample-size cautions.

NIST: Generalized ESD for several outliers

`https://www.itl.nist.gov/div898/handbook/eda/section3/eda35h3.htm`